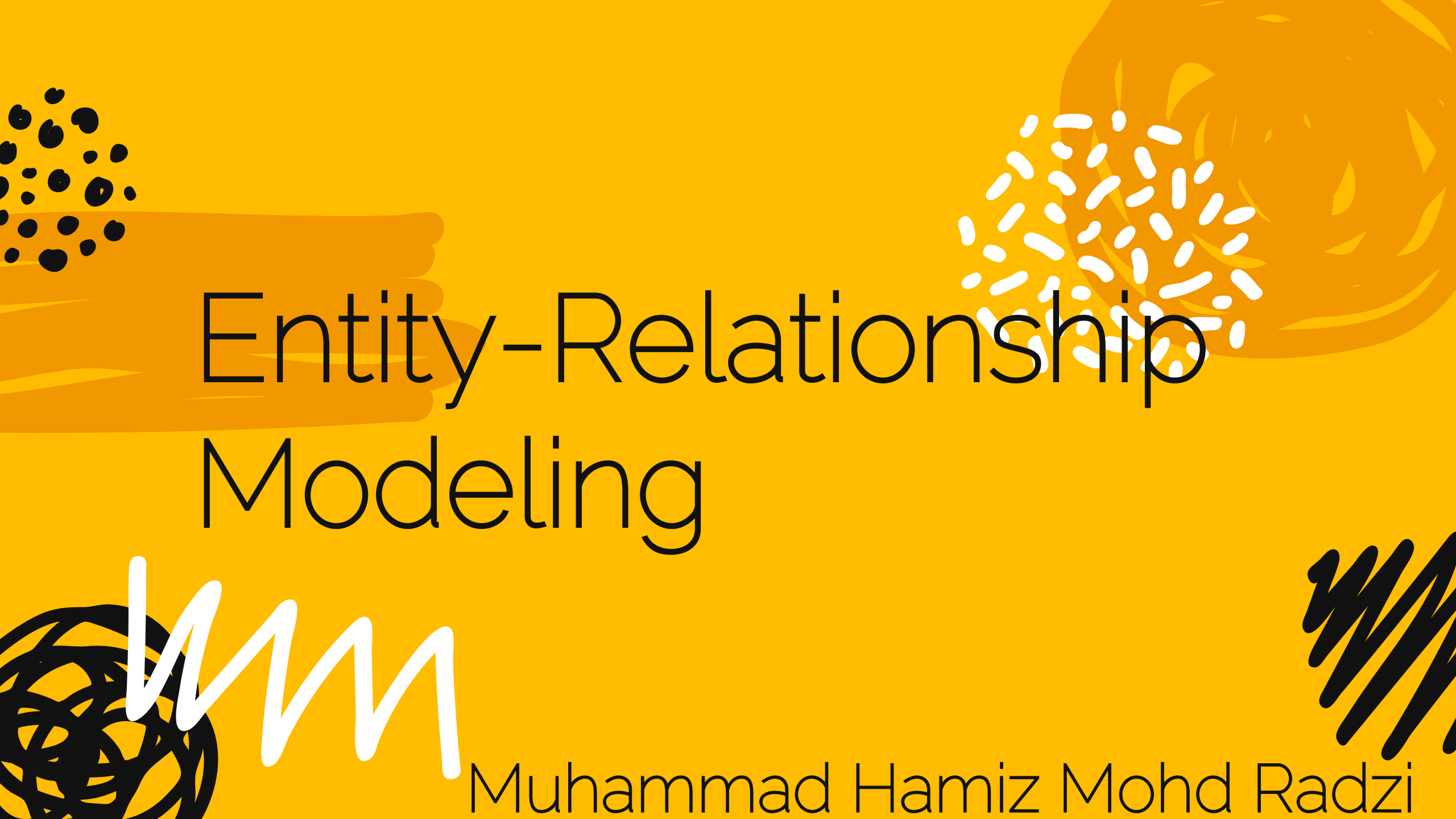




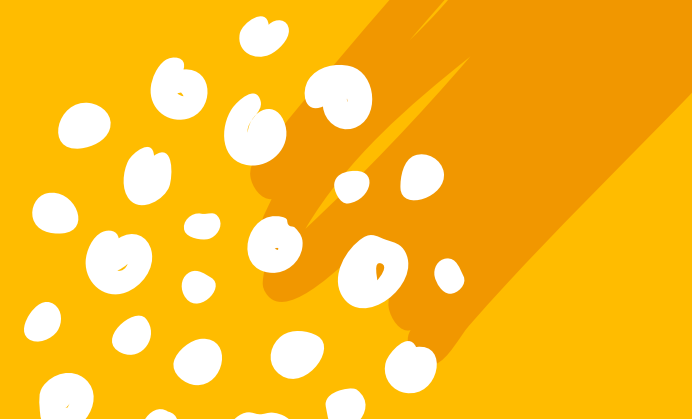
Entity-Relationship Modeling

Muhammad Hamiz Mohd Radzi



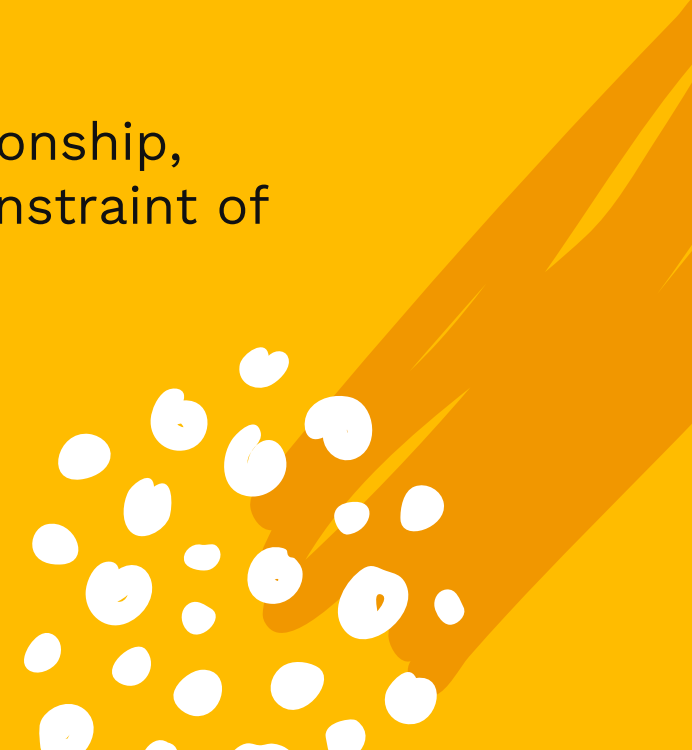
Objective

At the end of this lesson, you should be able to:

- X Define entity, entity types and entity occurrence
 - X Describe relationship types, occurrence, degree of relationship types, and recursive relationship.
 - X Explain attribute, attribute domain, simple, composite, single, multivalued and derived attributes.
 - X Describe the strong and weak entities.
 - X Explain attribute on relationship
- 



Objectives

- X Explain the multiplicity, cardinality and participation
 - X Describe fan and chasm trap
 - X Define the enhance ER Modeling
 - X Explain the concept of superclass/subclass, its relationship, specialization and generalization process, and the constraint of specialization/generalization.
- 



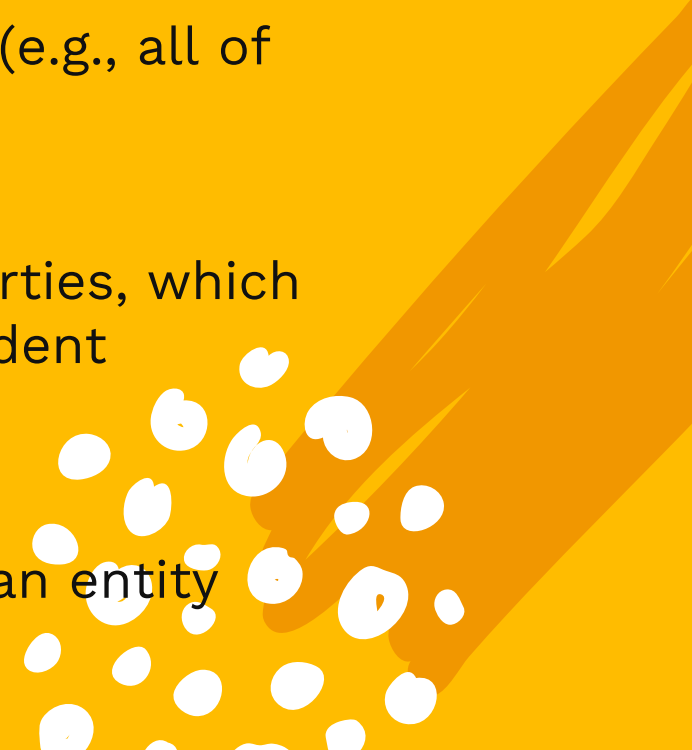
The Entity Relationship Model

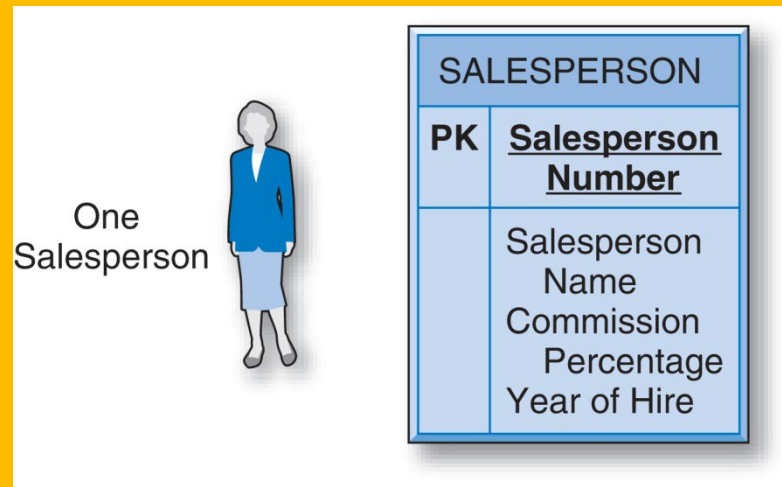
- X ERM is existed as to explore the different ways that entities can relate to each other as they always do in the real world.
 - X ERM also is used to devise a way of recording, of diagramming, the entities and the ways in which they interrelate in the business environment.
 - X The things that you need to know in order to construct the ERM are:
 - X Entity
 - X Attribute
 - X Relationship
- 



Entity



- X Entity - a “thing” or “object” in our environment that we want to keep track of.
 - X Entity set - A collection of entities of the same type (e.g., all of the company’s employees).
 - X Entity type - A group of objects with the same properties, which are identified by the enterprise as having an independent existence.
 - X Entity occurrence - A uniquely identifiable object of an entity type.
- 



- X Entity is in rectangular shape
- X Name of entity is in caps above the separator line.
- X For example:
Entity type = SALESPERSON



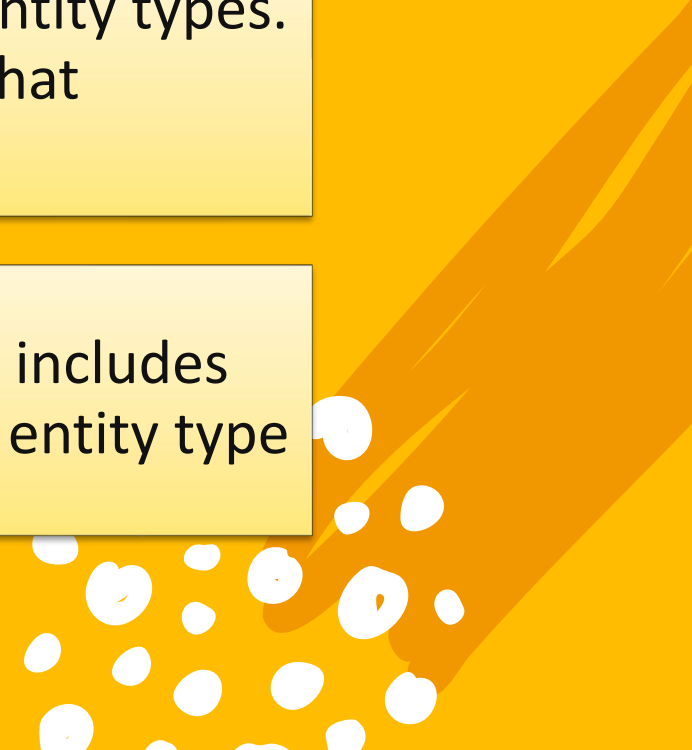
Relationship Types



Relationship Type

- Set of meaningful associations among entity types. Each relationship type is given a name that describes its function.

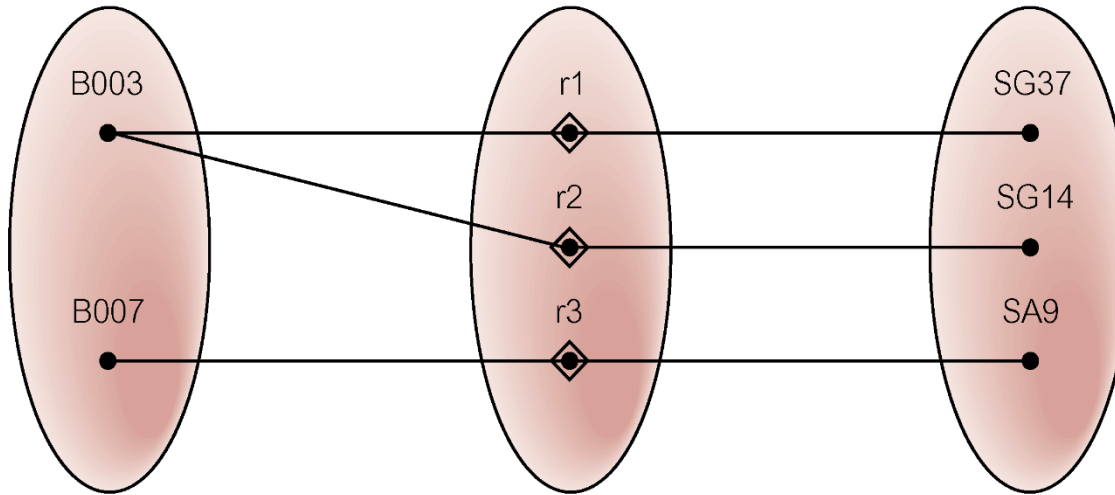
Relationship occurrence

- Uniquely identifiable association, which includes one occurrence from each participating entity type
- 

Branch entity
(branchNo)

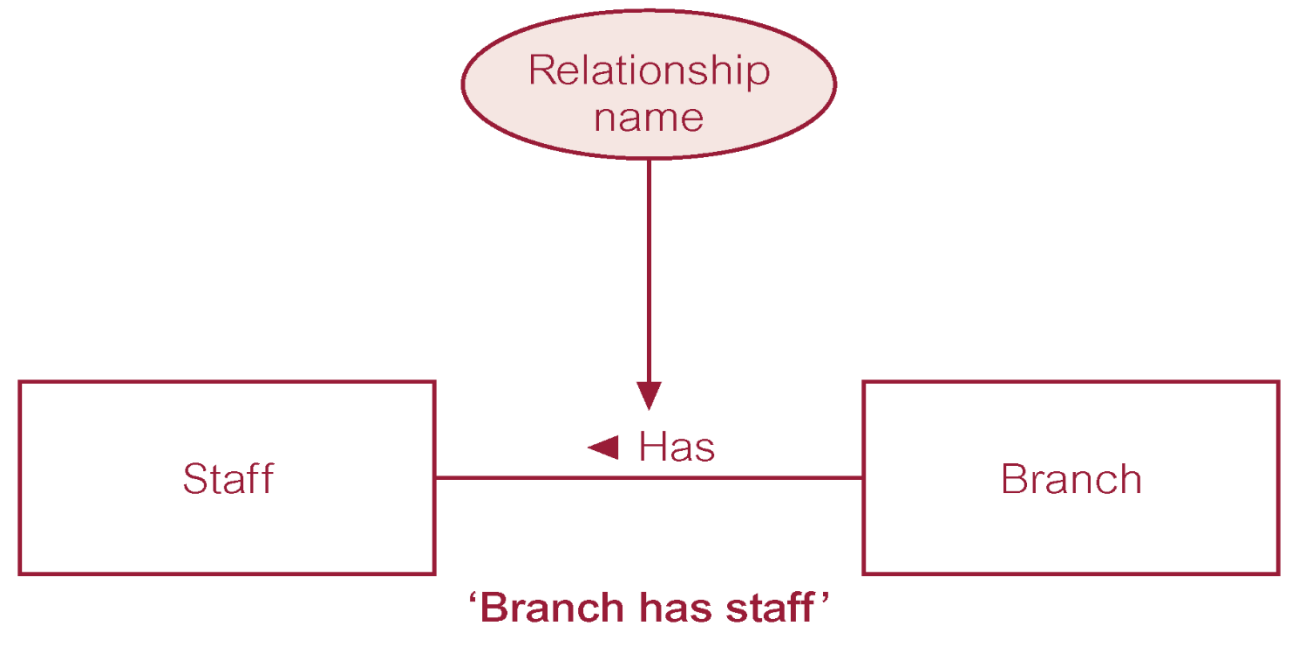
Has ►
relationship

Staff entity
(staffNo)



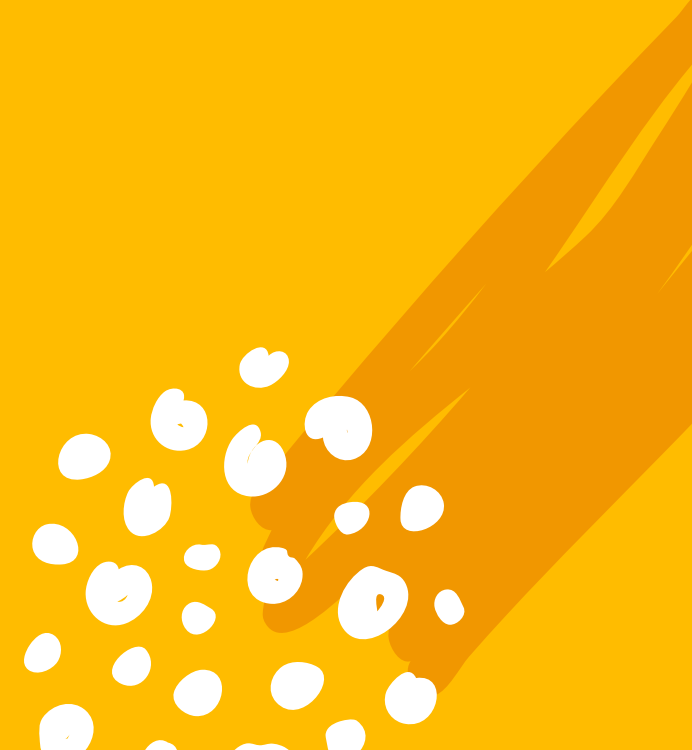
Semantic Net View

Diagrammatic View

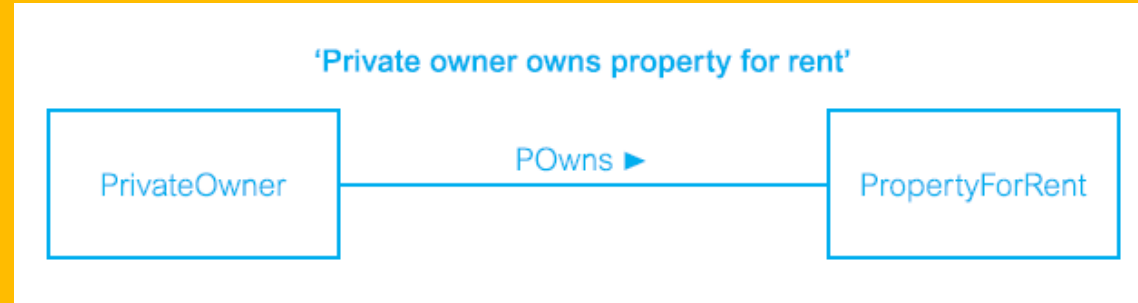




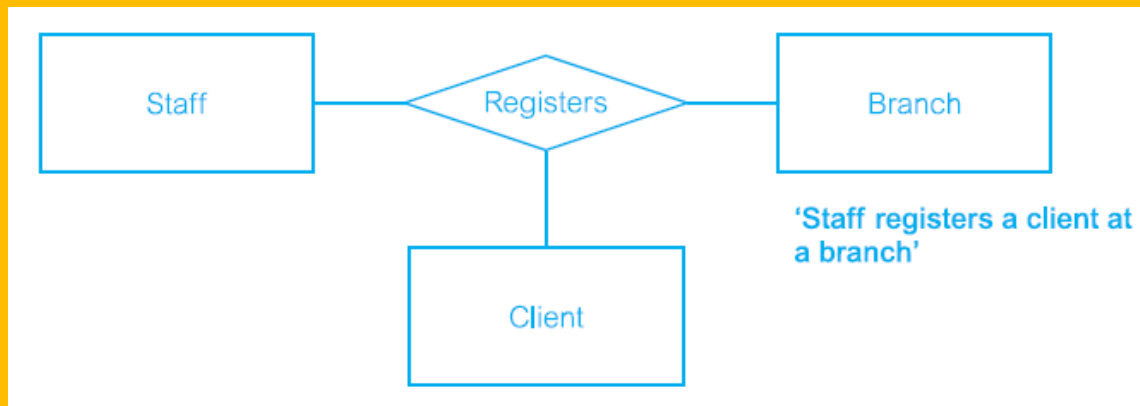
Degree of Relationship

- x It is defined as number of participating entities in relationship.
 - x Relationship of degree :
 - x two is binary
 - x three is ternary
 - x four is quaternary.
- 
- 

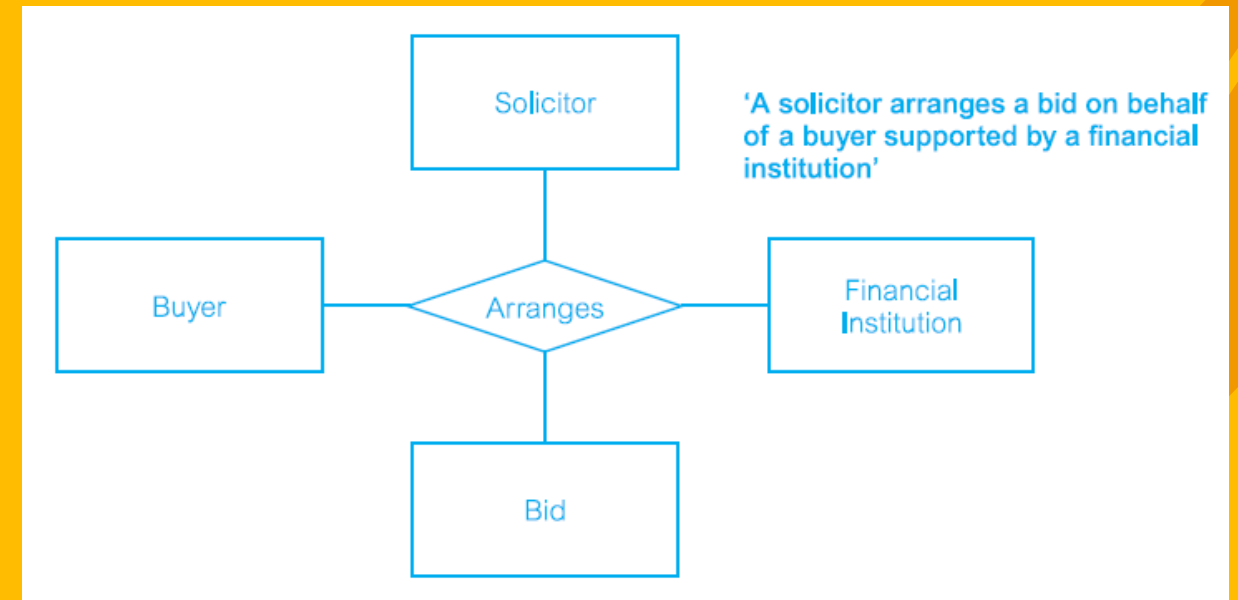
Binary Relationship



Ternary Relationship

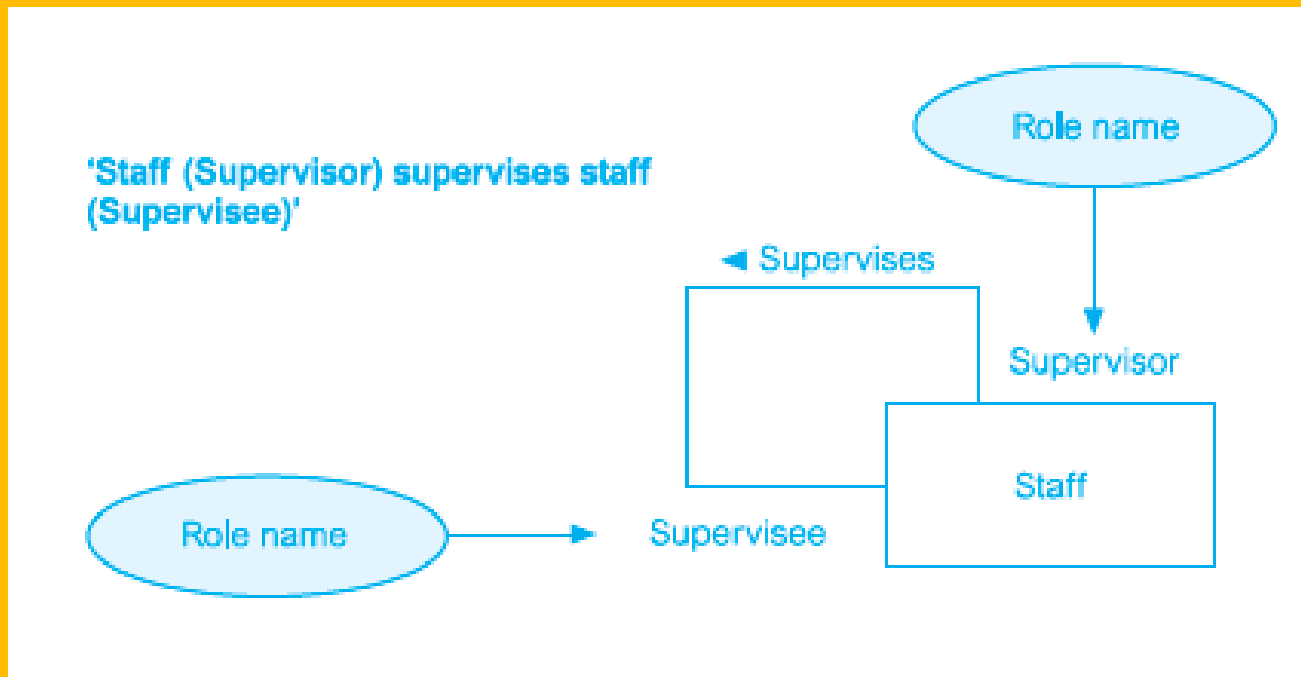


Quaternary Relationship

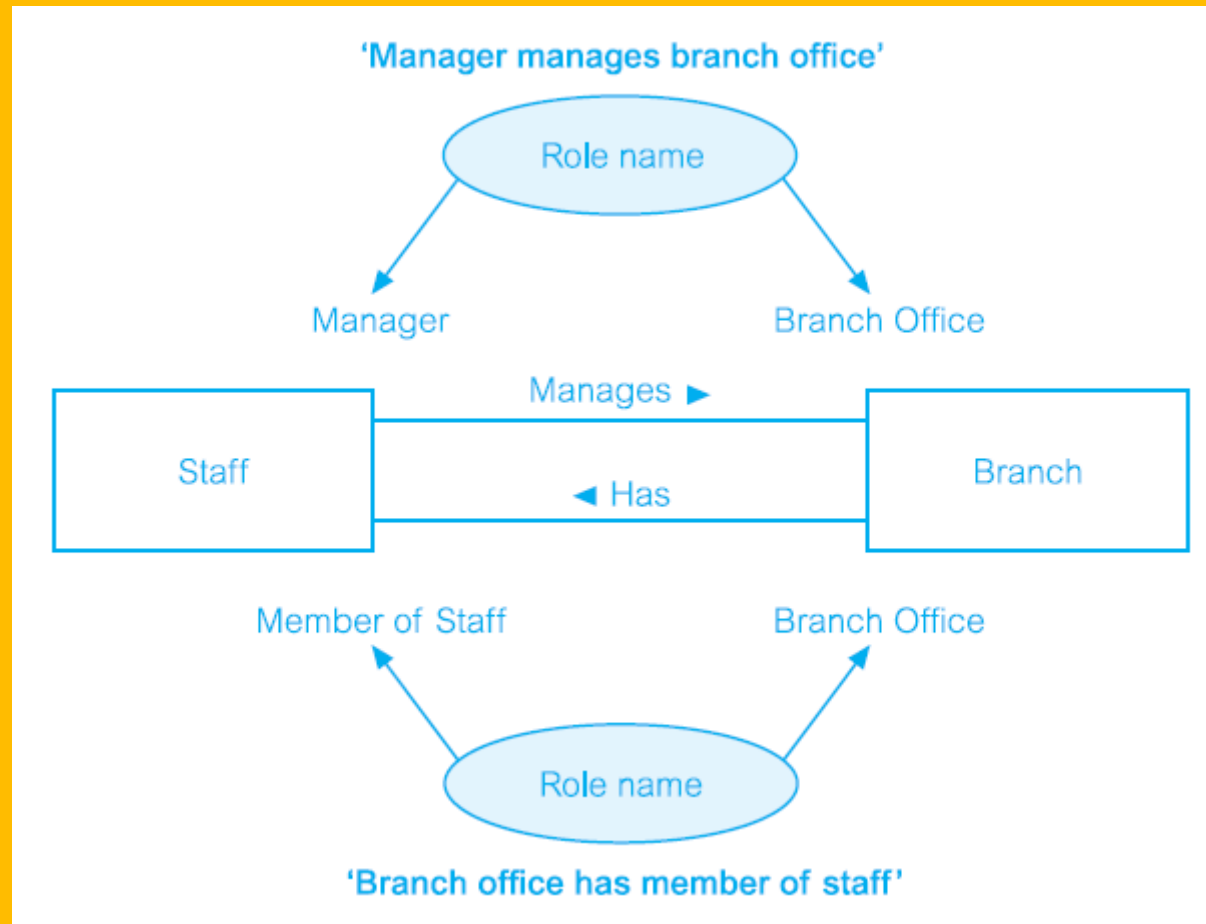


Recursive Relationship

- x Recursive relationship - A relationship type where the same entity type participates more than once in different roles.



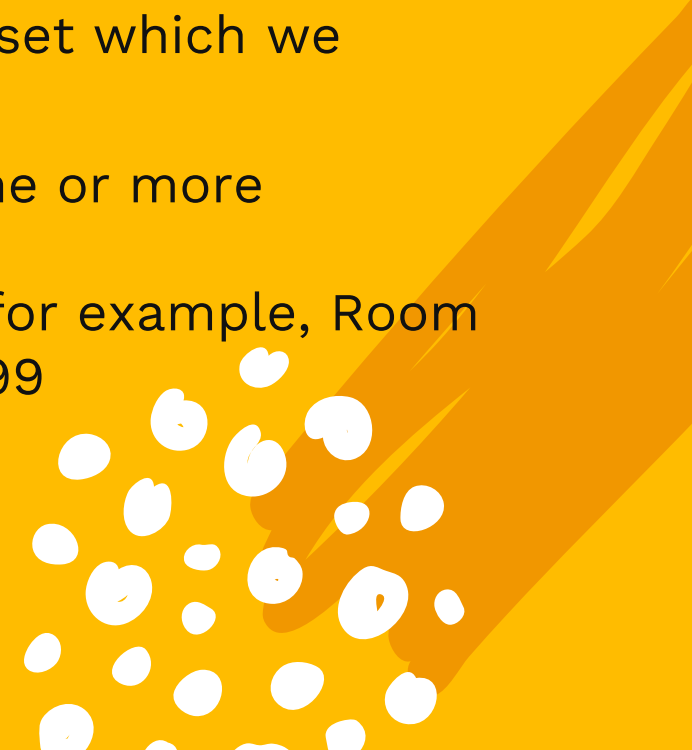
Entities associated through two distinct relationships with role names

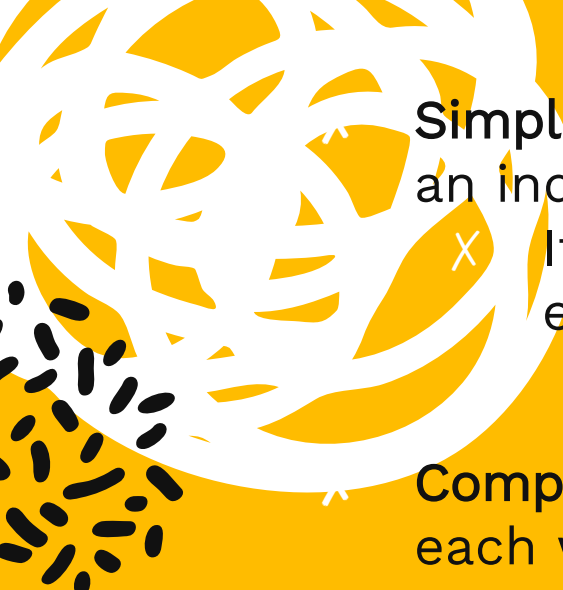




Attributes



- X Attribute - a property of, a characteristic of, or a fact that we know about an entity.
 - X Some attributes have unique values within an entity set which we called as Primary Key (PK)
 - X Attribute domain - The set of allowable values for one or more attributes.
 - X Each attributes have its own domain of values, for example, Room Number must be an Integer range from 1000-1999
- 



Simple attribute - An attribute composed of a **single component** with an independent existence.

X It cannot be further subdivided into smaller components. For example, Salary.

Composite attribute - An attribute composed of **multiple components**, each with an independent existence.

X Some attributes can be further divided to yield smaller components. For example, address can be divided into Street, City, Postcode.

X **Single-valued attribute** - An attribute that holds a **single value** for each occurrence of an entity type.

X For example, Salesperson Number. Only one value can be attached to the attribute.

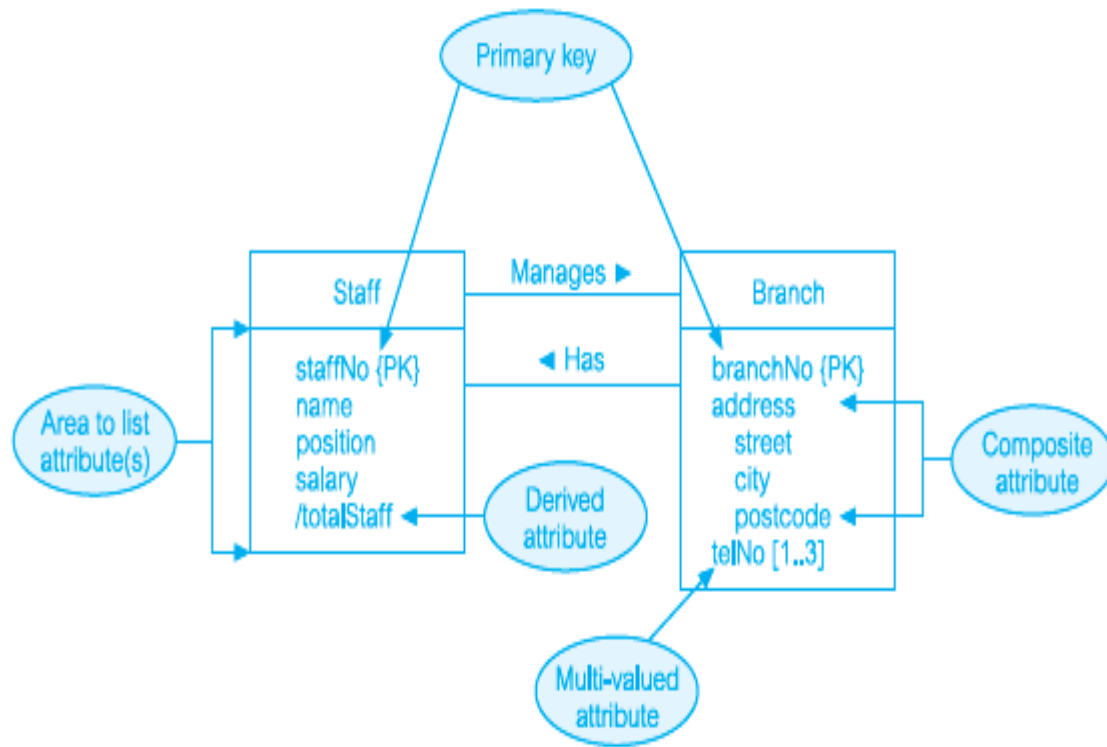


X **Multi-valued attribute** - An attribute that holds **multiple values** for each occurrence of an entity type.

X For example, branch number B003 has telephone numbers 0141-339-2178 and 0141-339-4439

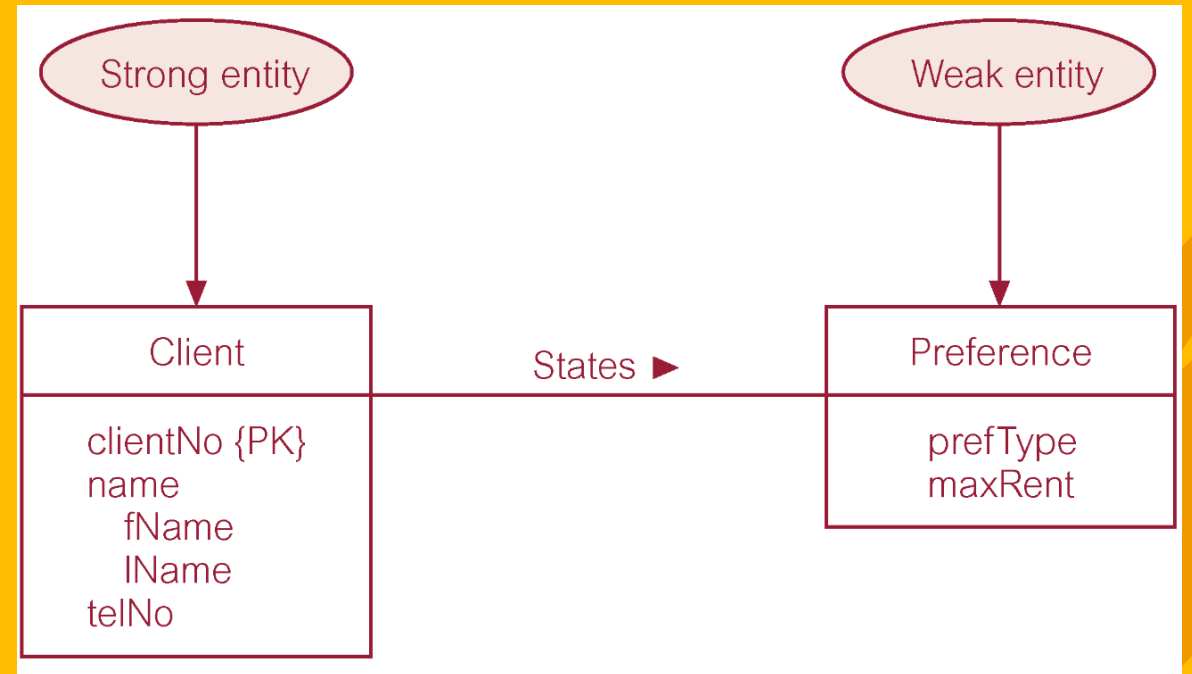
X **Derived attribute** - An attribute that represents a value that is **derivable from the value of a related attribute** or set of attributes, not necessarily in the same entity type.

X For example, the total number of staff (totalStaff) attribute of the Staff entity type can be calculated by counting the total number of Staff entity occurrences.

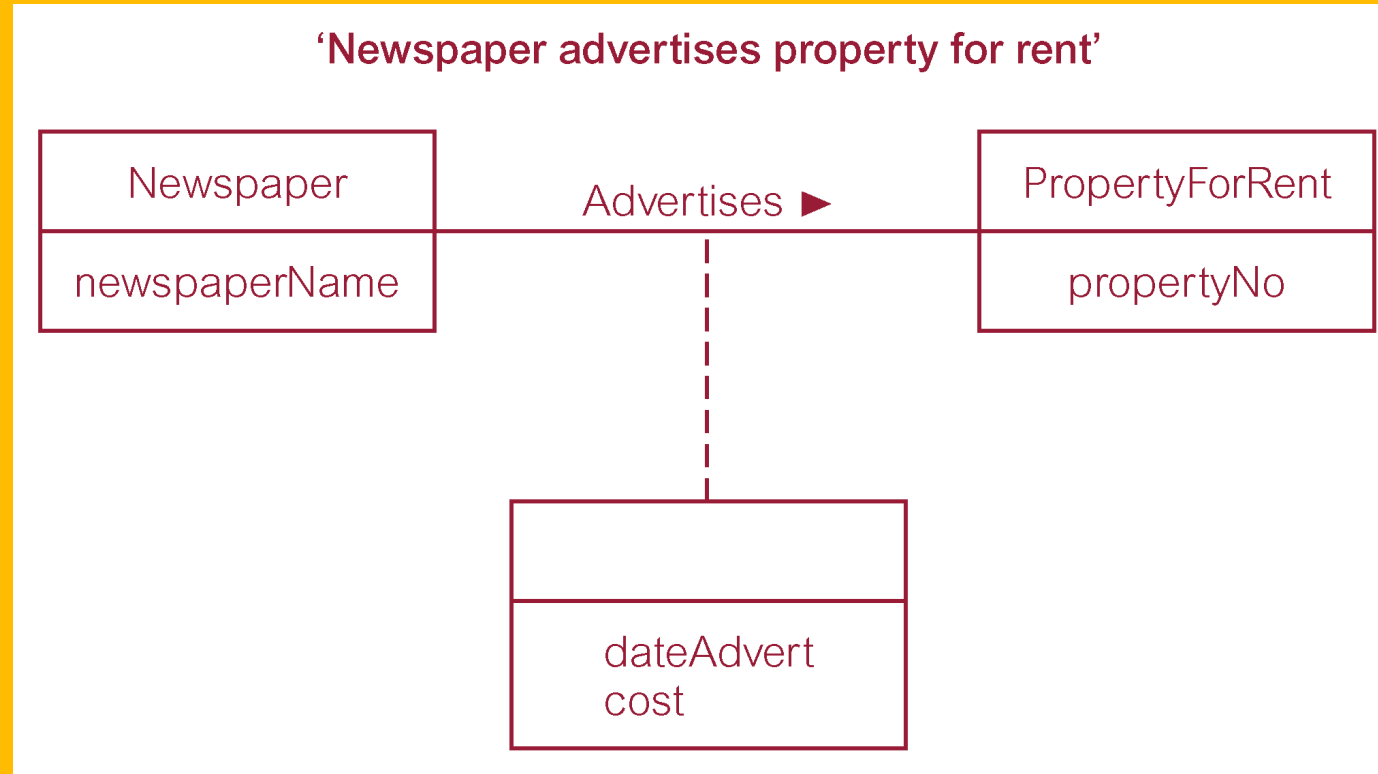


Strong and Weak Entity Types

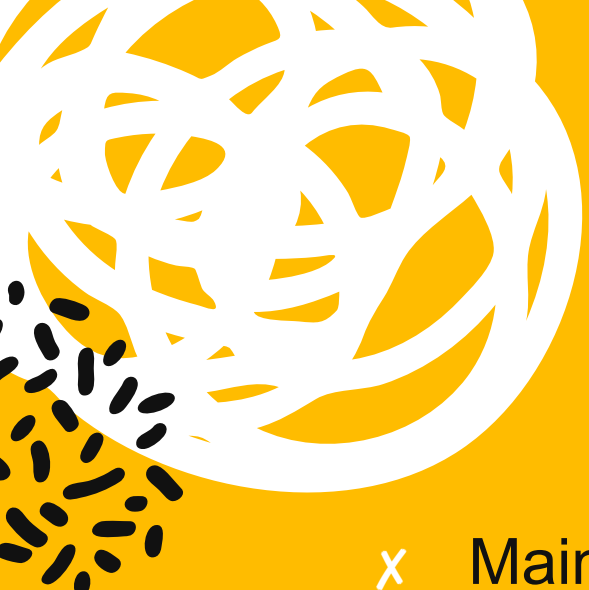
- x **Strong entity type:** An entity type that is *not* existence-dependent on some other entity type.
- x **Weak entity type:** An entity type that is existence-dependent on some other entity type.



Attributes on Relationship



x Later will be converted as bridge entity



Structural Constraints

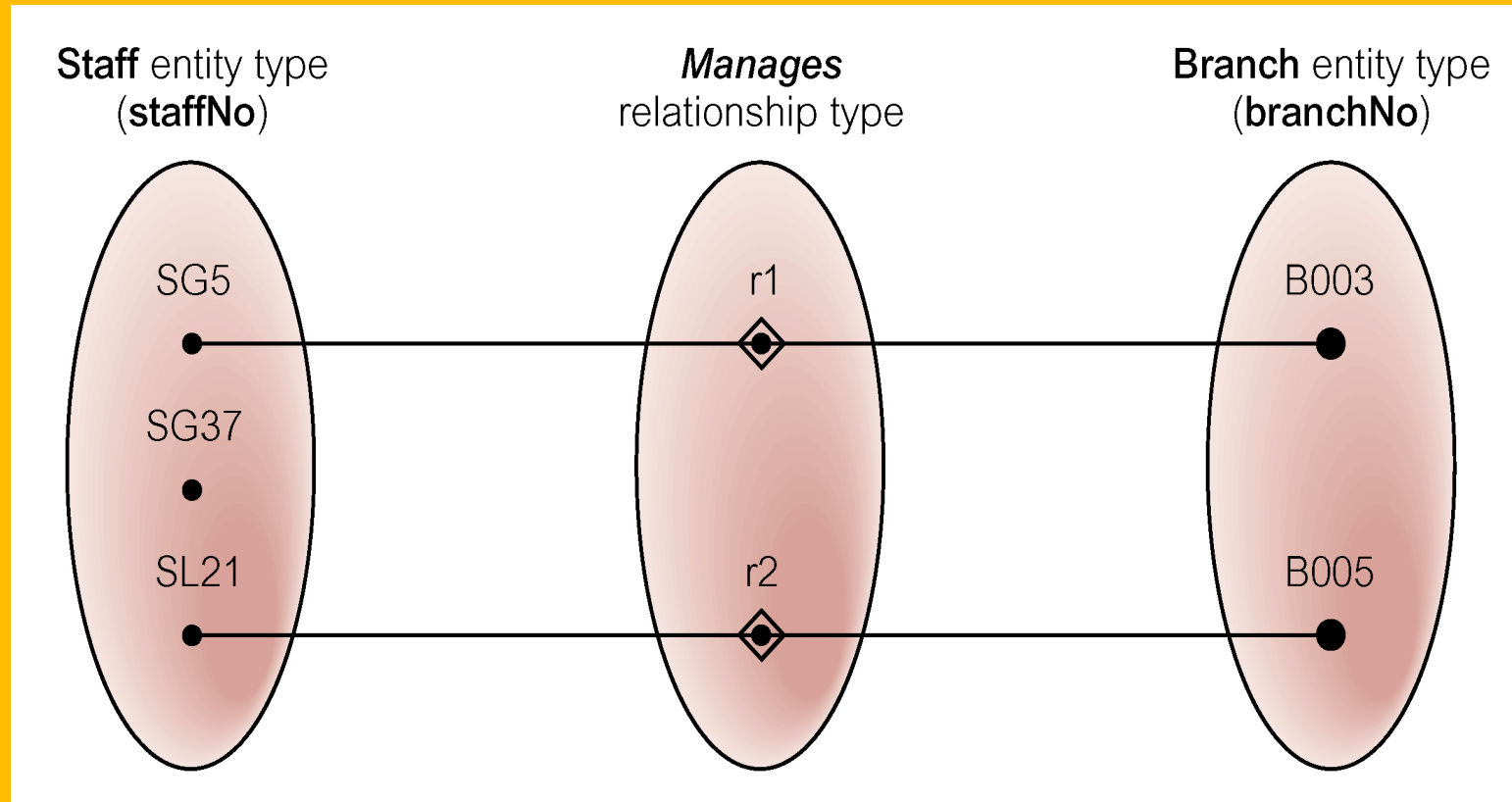
- x Main type of constraint on relationships is called **multiplicity**.
 - x **Multiplicity** - number (or range) of possible occurrences of an entity type that may relate to a single occurrence of an associated entity type through a particular relationship.
 - x Represents policies (called *business rules*) established by user or company.
 - x The most common degree for relationships is binary.
- 



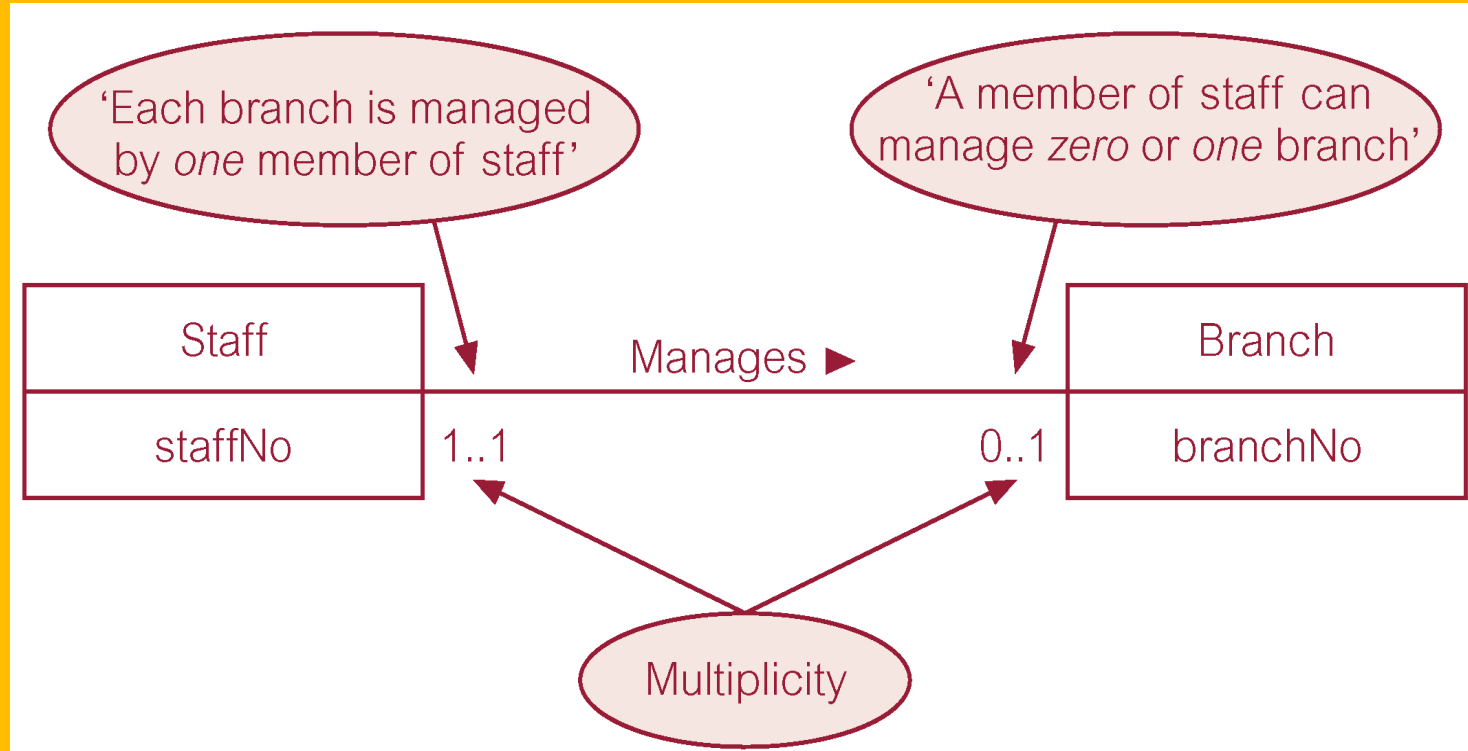
x Binary relationships are generally referred to as being:

- x one-to-one (1:1)
 - x one-to-many (1:*)
 - x many-to-many (*:*)
- 

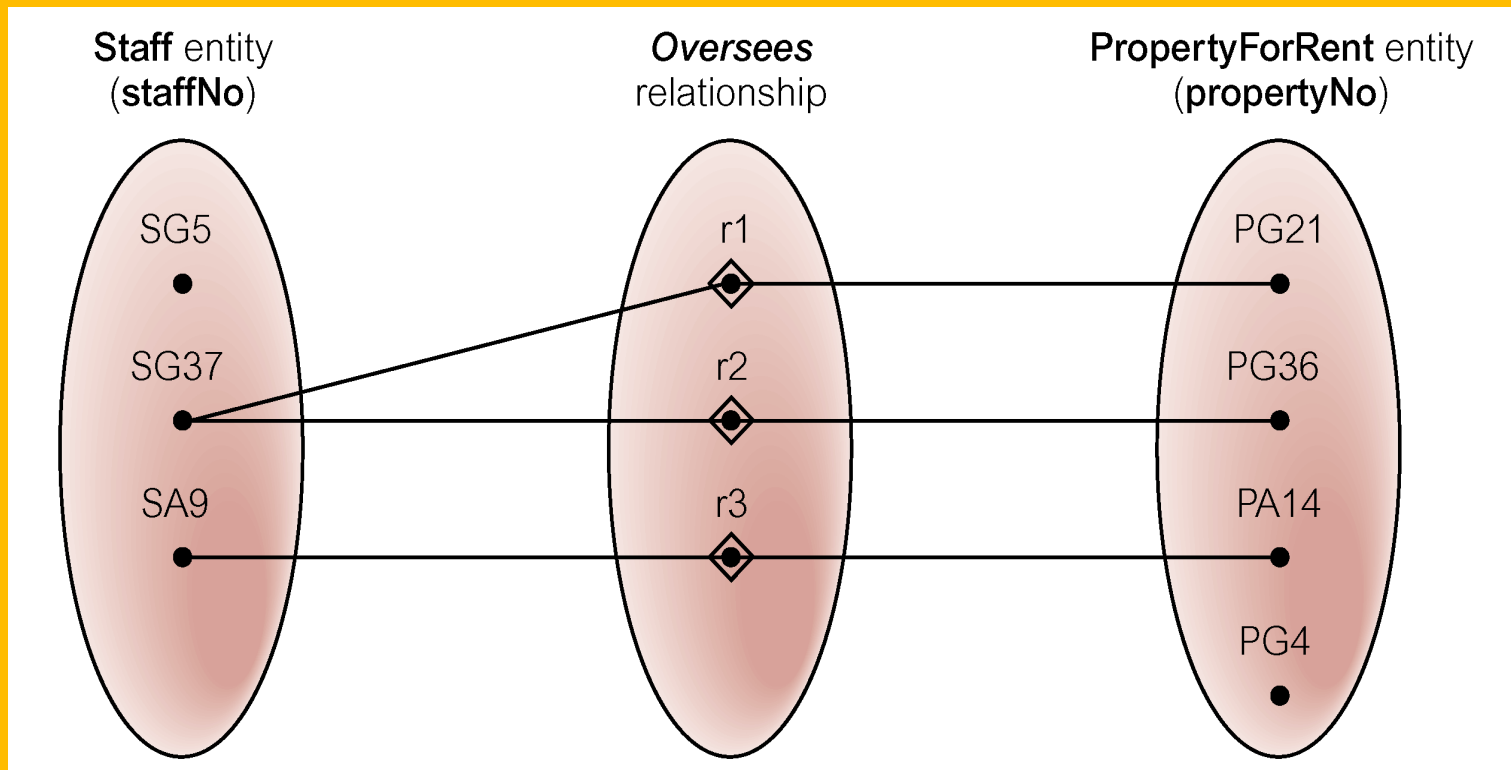
Semantic net of Staff *Manages* Branch relationship type



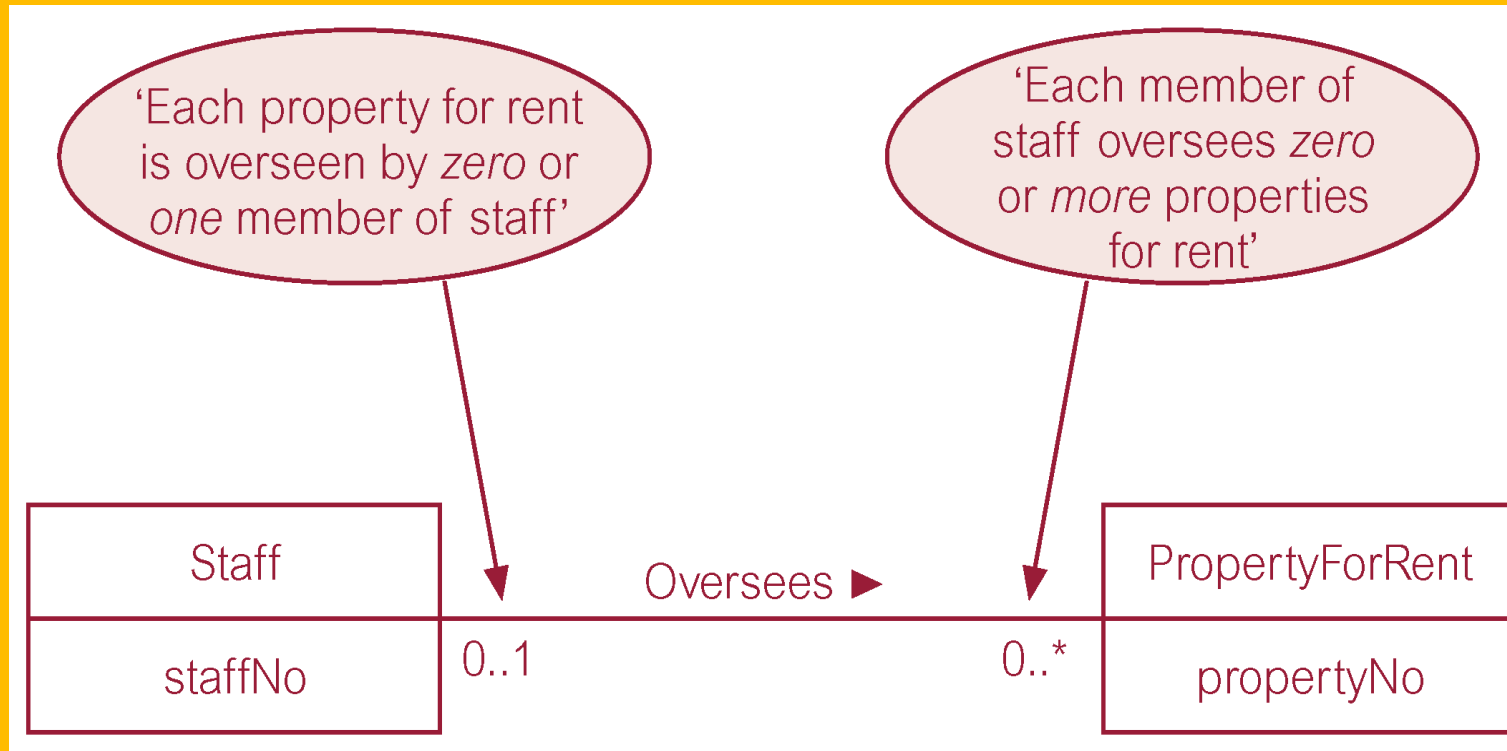
Multiplicity of Staff *Manages* Branch (1:1) relationship



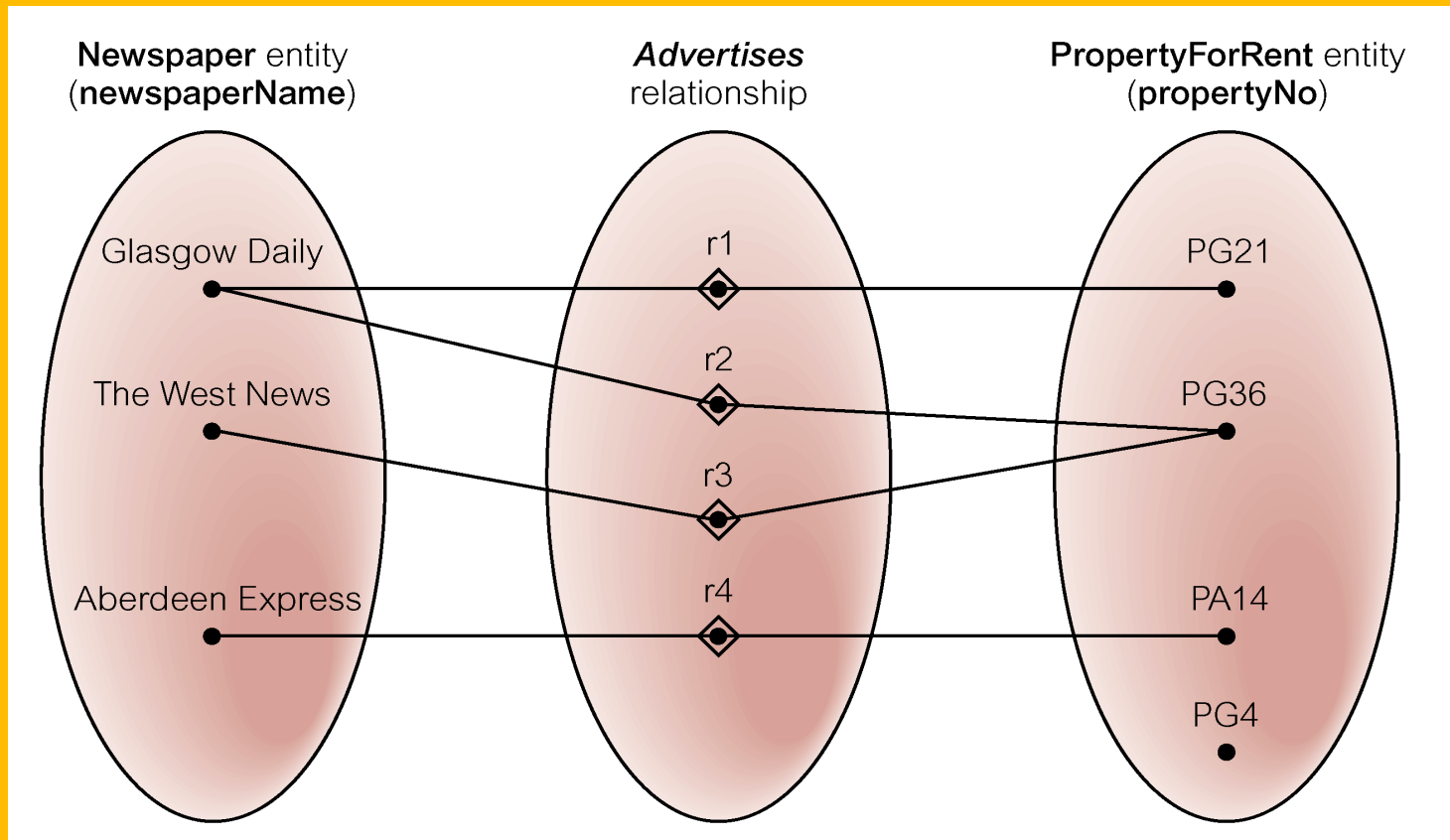
Semantic net of Staff Oversees PropertyForRent relationship type



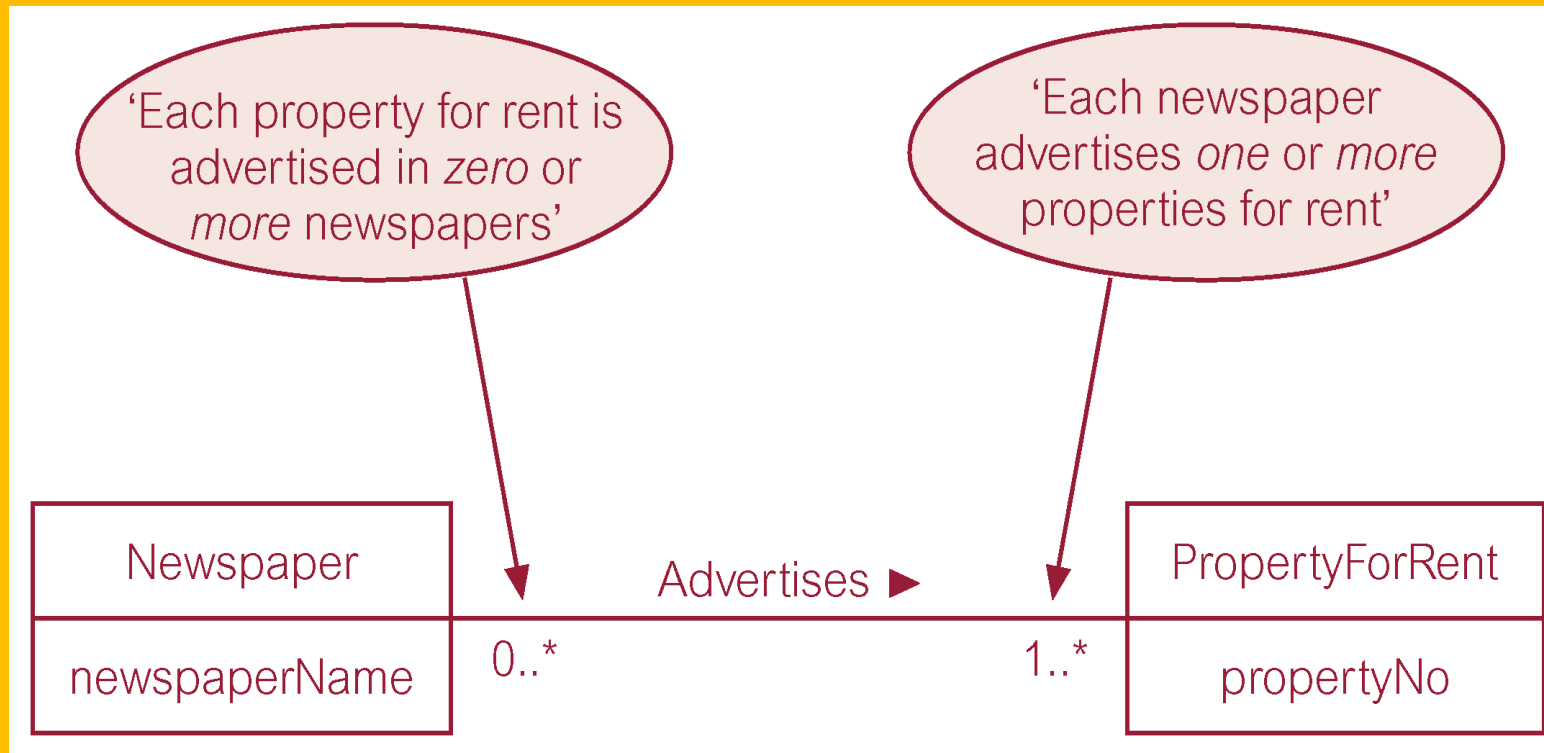
Multiplicity of Staff *Oversees* PropertyForRent (1:*)
relationship type



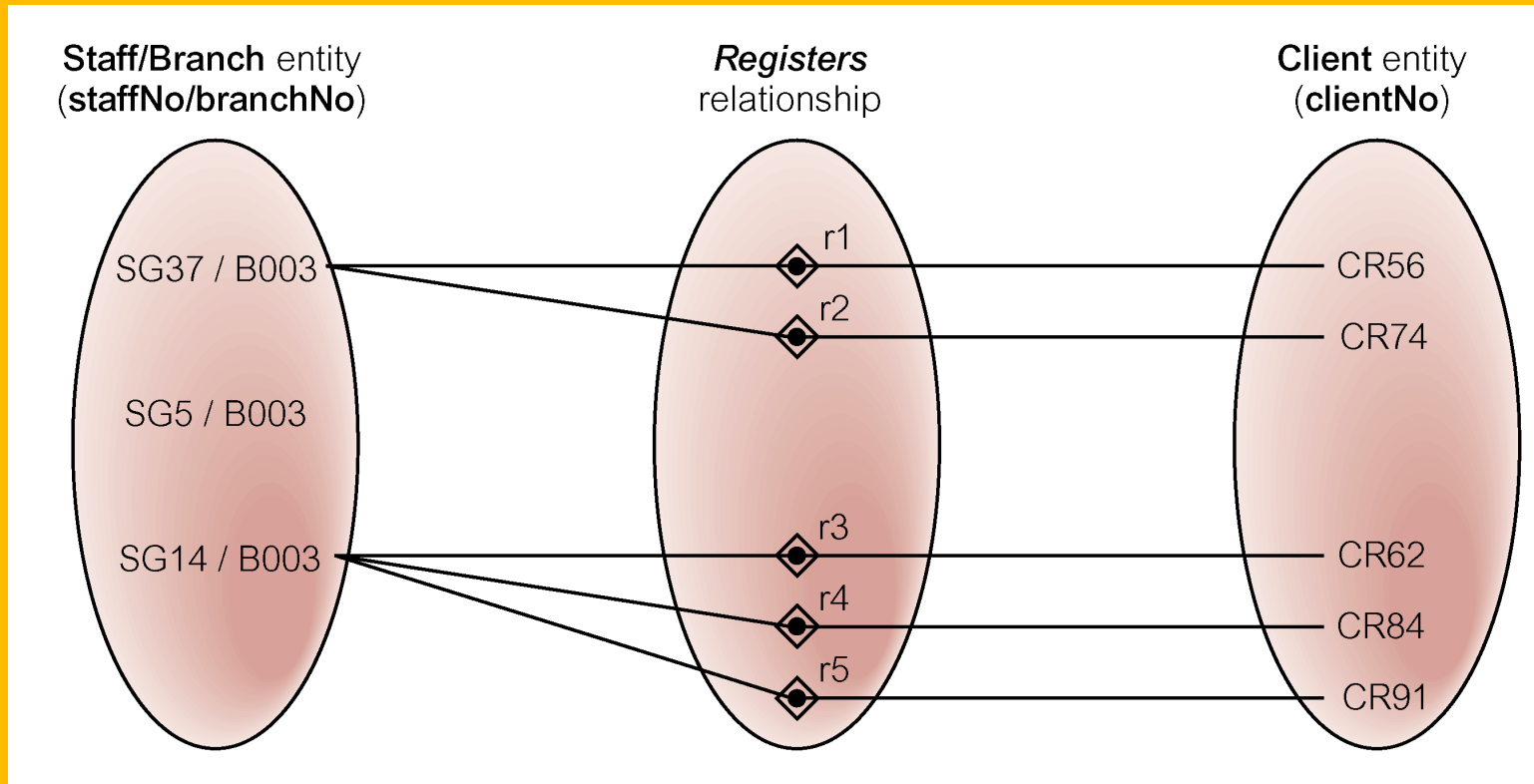
Semantic net of Newspaper *Advertises* PropertyForRent relationship type



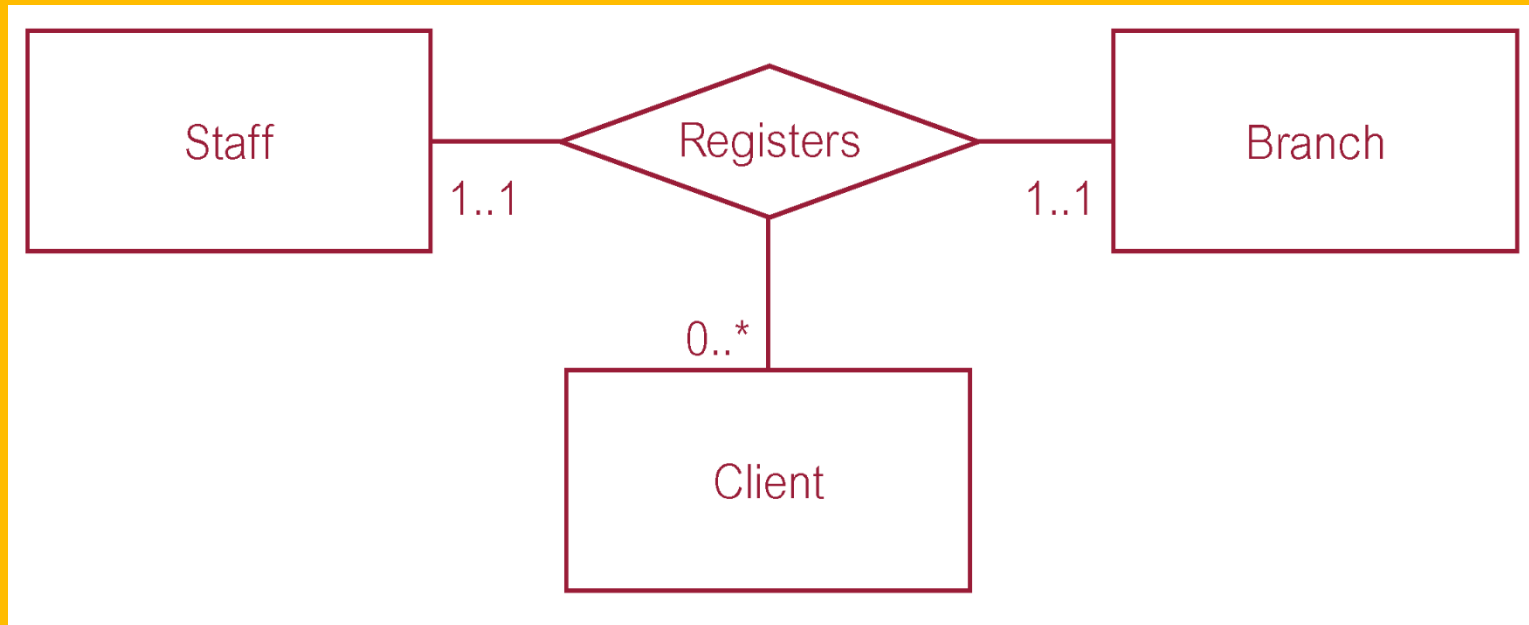
Multiplicity of Newspaper *Advertises* PropertyForRent (*:*) relationship



Semantic net of ternary *Registers* relationship with values for Staff and Branch entities fixed



Multiplicity of ternary *Registers* relationship



Summary of Multiplicity Constraints

Alternative ways to represent
multiplicity constraints

Meaning

0..1

Zero or one entity occurrence

1..1 (or just 1)

Exactly one entity occurrence

0..* (or just *)

Zero or many entity occurrences

1..*

One or many entity occurrences

5..10

Minimum of 5 up to a maximum of 10 entity occurrences

0, 3, 6–8

Zero or three or six, seven, or eight entity occurrences

Structural Constraints

- x Multiplicity is made up of two types of restrictions on relationships: *cardinality* and *participation*.

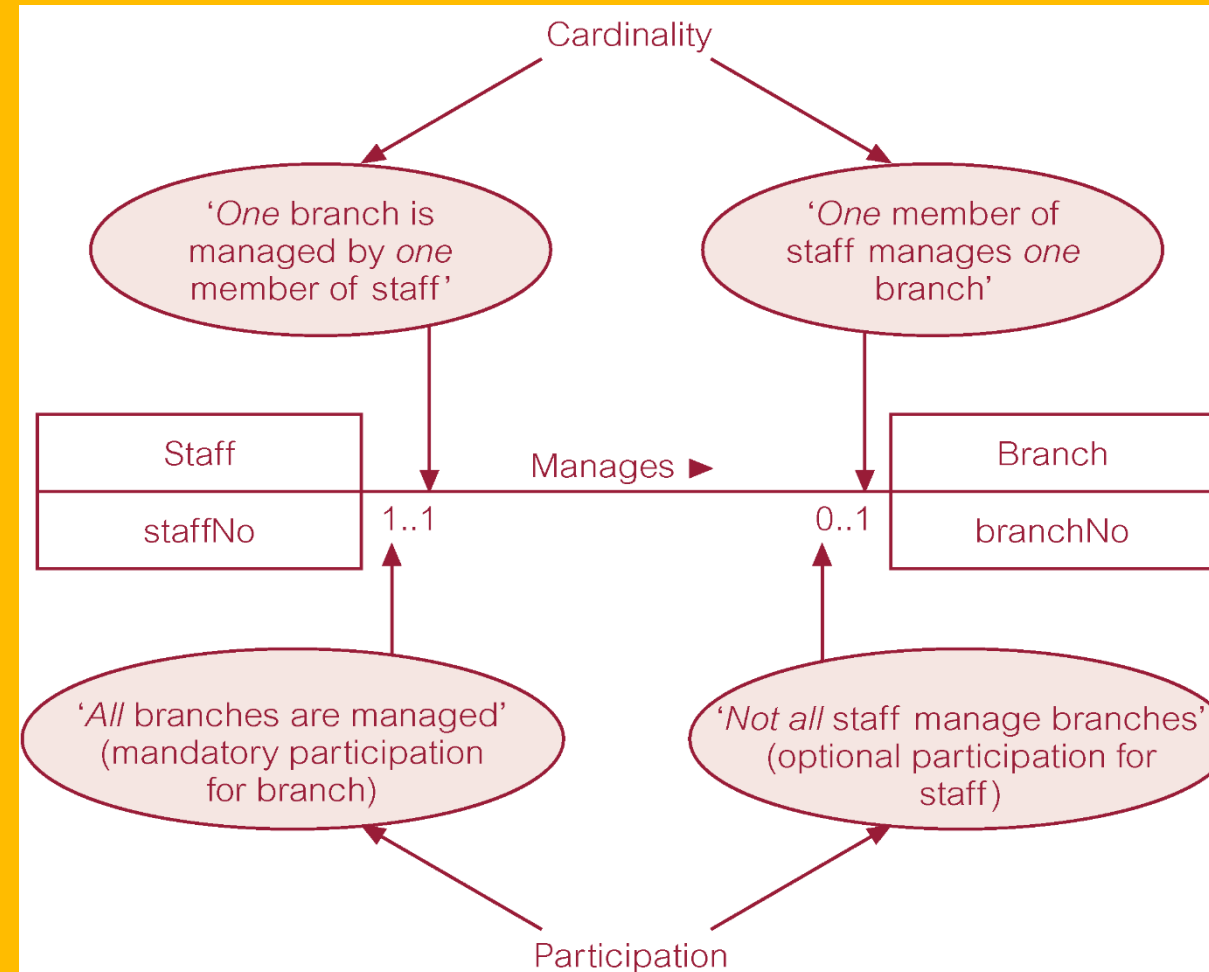
Cardinality

- Describes maximum number of possible relationship occurrences for an entity participating in a given relationship type.

Participation

- Determines whether all or only some entity occurrences participate in a relationship.

Multiplicity as cardinality and participation constraints

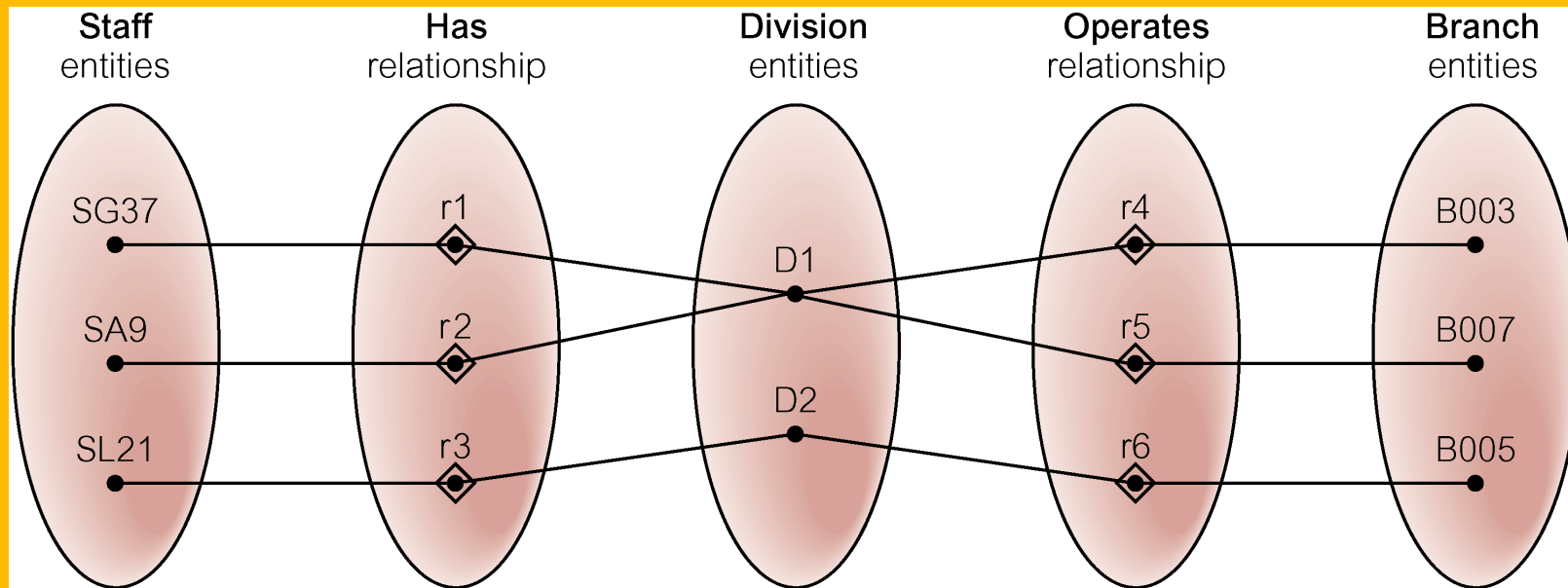




Problems with ER Modeling

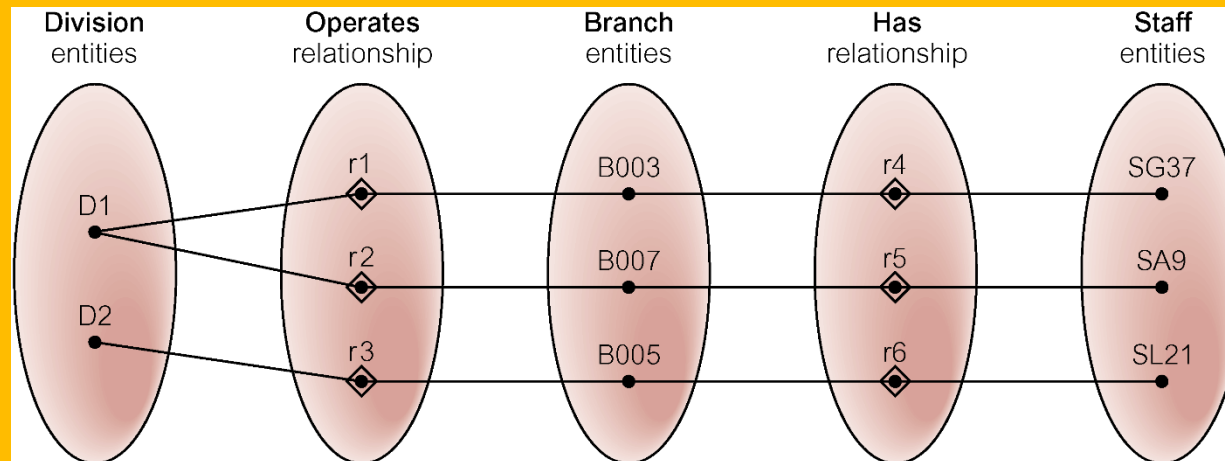
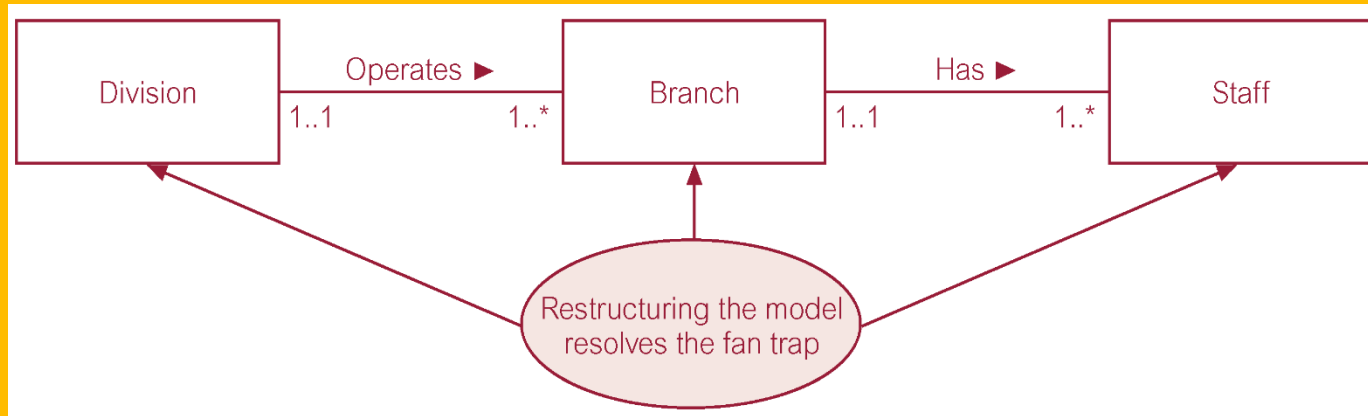
- X Problems may arise when designing a conceptual data model called connection traps.
 - X Often due to a misinterpretation of the meaning of certain relationships.
 - X Fan Traps: Where a model represents a relationship between entity types, **but pathway between certain entity occurrences is ambiguous.**
 - X Chasm Trap: Where a model suggests the existence of a relationship between entity types, but **pathway does not exist** between certain entity occurrences.
- 

Fan Trap



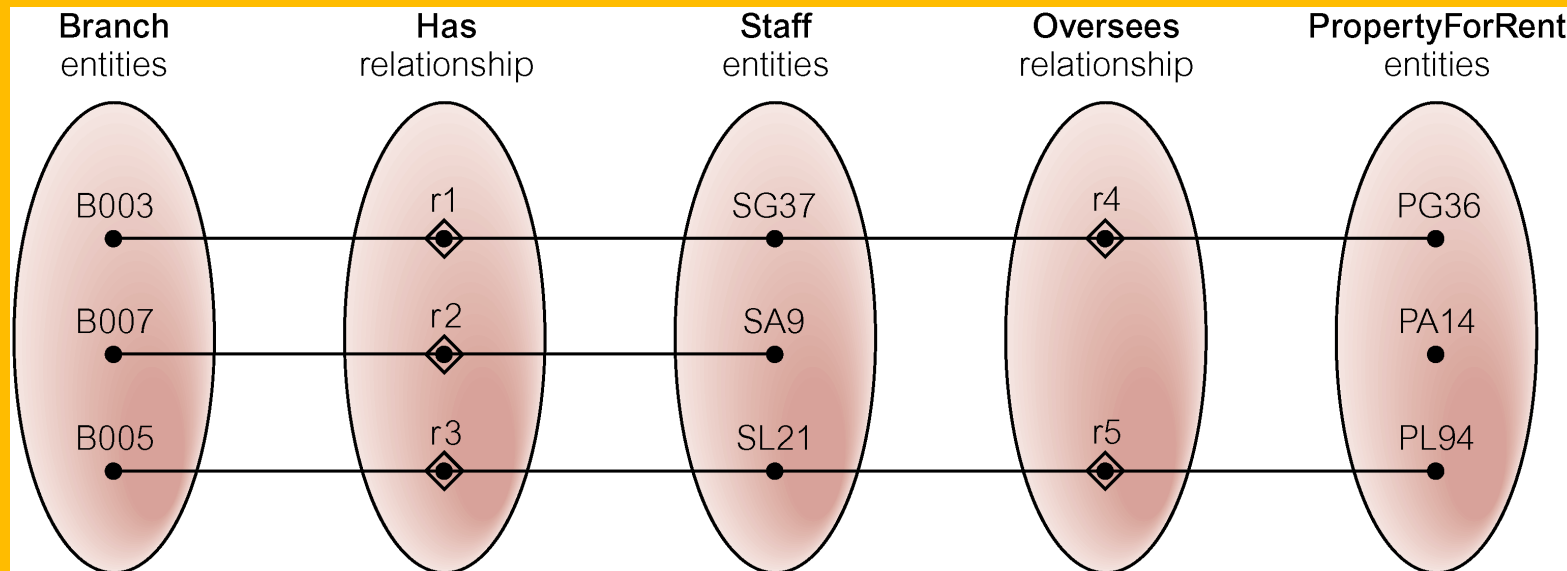
x At which branch office does staff number SG37 work?

Solution of Fan Trap



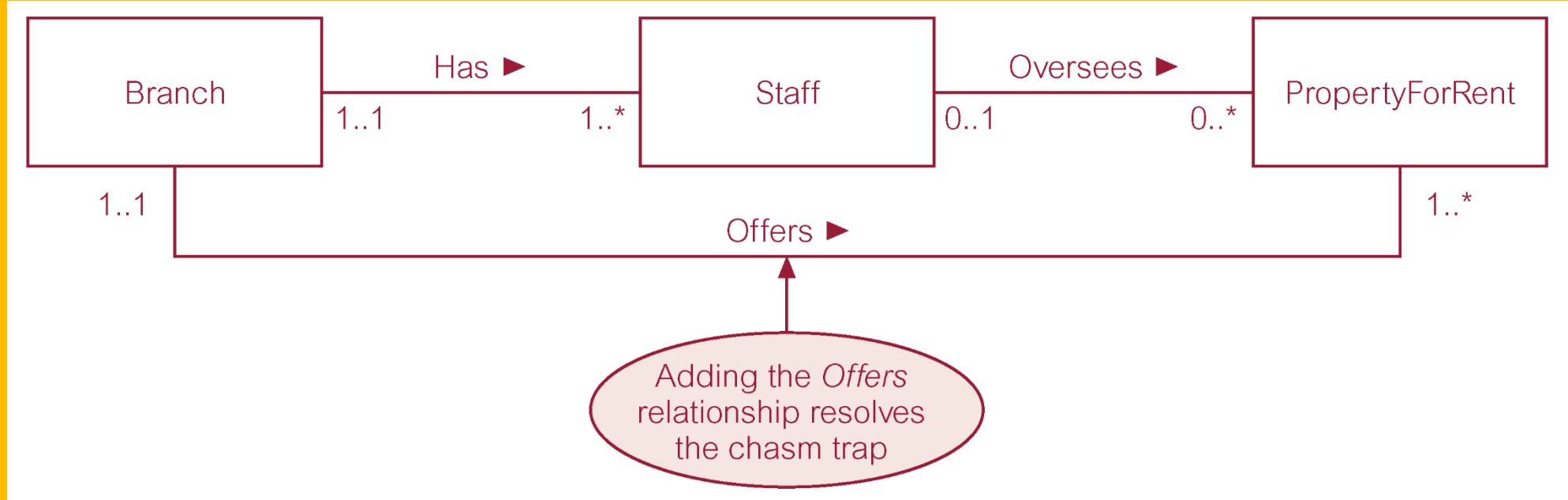
X SG37 works at branch B003.

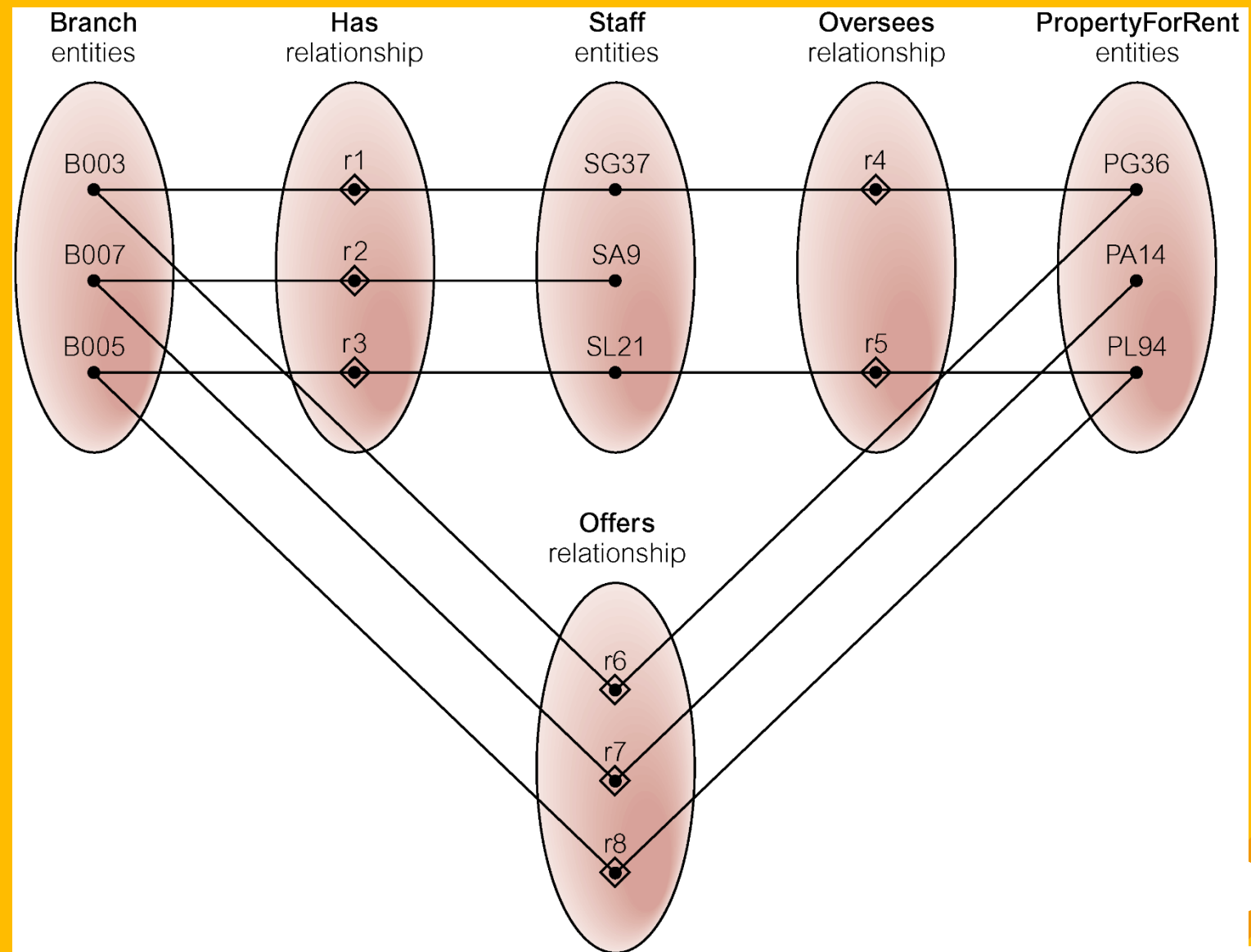
Chasm Trap



x At which branch office is property PA14 available?

Solution of Chasm Trap





x PA14 is available at B007 branch

The Enhanced Entity-Relationship Model



Since 1980s there has been an increase in emergence of new database applications with more demanding requirements.



Basic concepts of ER modeling are not sufficient to represent requirements of newer, more complex applications.



Response is development of additional 'semantic' modeling concepts.



Semantic concepts are incorporated into the original ER model and called the Enhanced Entity-Relationship (EER) model.



Examples of additional concept of EER model is called specialization / generalization.



Specialization / Generalization



Superclass

- An entity type that includes one or more distinct subgroupings of its occurrences.

Subclass

- A distinct subgrouping of occurrences of an entity type.

- x Superclass/subclass relationship is one-to-one (1:1).
 - x Superclass may contain overlapping or distinct subclasses.
 - x NOT ALL members of a superclass need be a member of a subclass.
- 

Specialization / Generalization

Attribute Inheritance

- An entity in a subclass represents same 'real world' object as in superclass, and may possess subclass-specific attributes, as well as those associated with the superclass.

Specialization

- Process of maximizing differences between members of an entity by identifying their distinguishing characteristics.

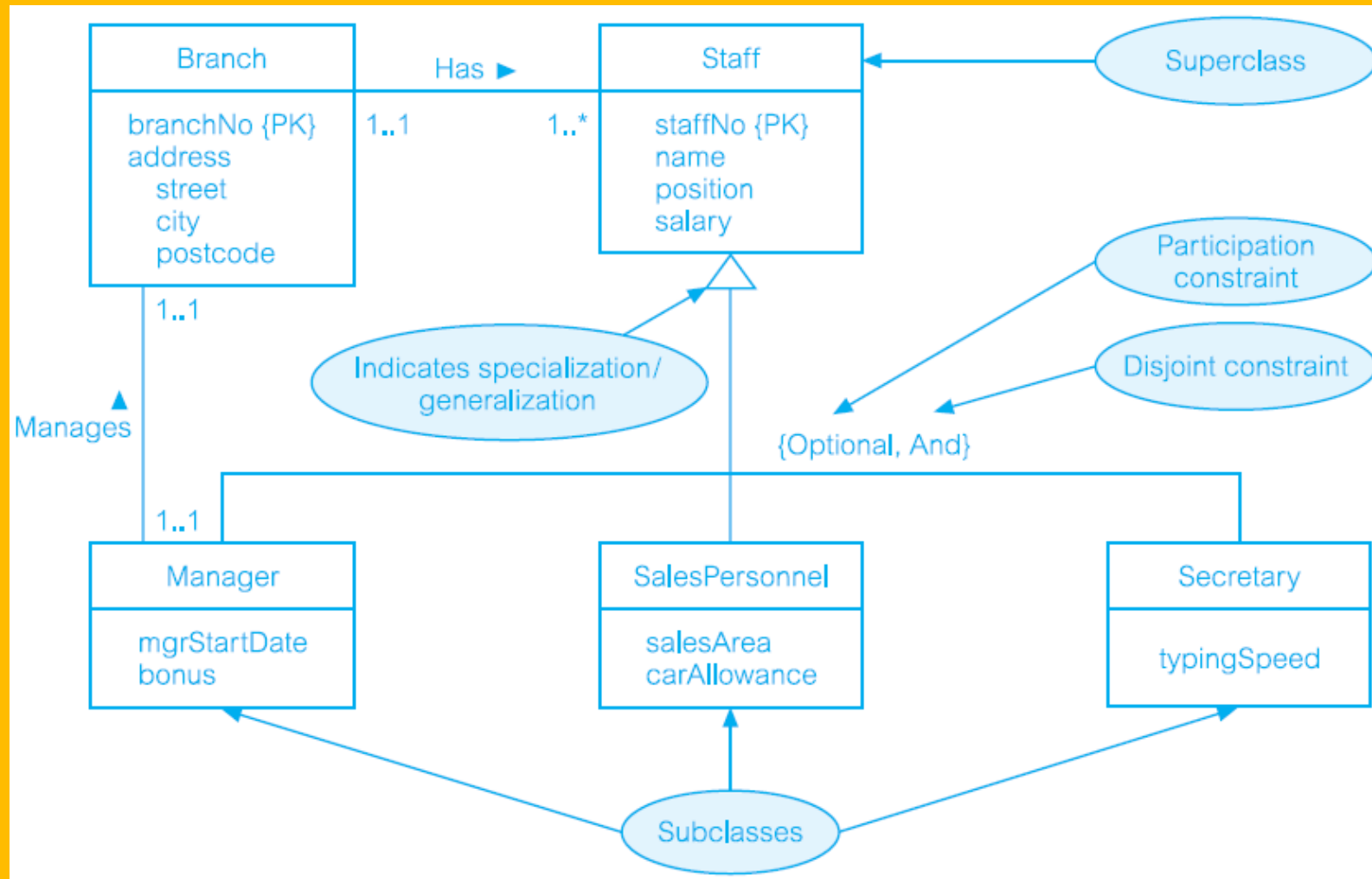
Generalization

- Process of minimizing differences between entities by identifying their common characteristics.

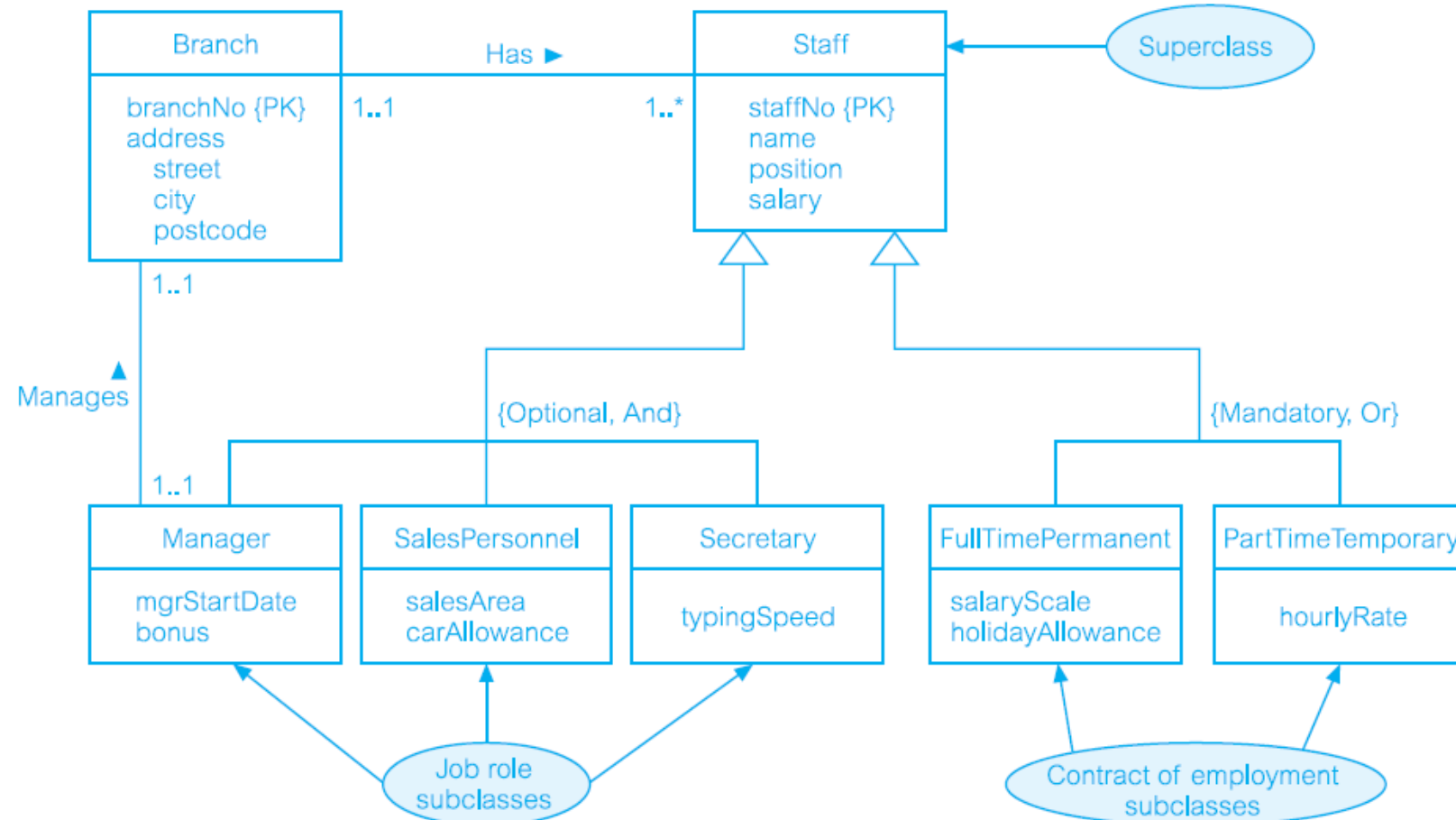
AllStaff relation holding details of all staff

Attributes appropriate for all staff				Attributes appropriate for branch Managers		Attributes appropriate for Sales Personnel		Attribute appropriate for Secretarial staff
staffNo	name	position	salary	mgrStartDate	bonus	sales Area	car Allowance	typing Speed
SL21	John White	Manager	30000	01/02/95	2000	SA1A	5000	100
SG37	Ann Beech	Assistant	12000					
SG66	Mary Martinez	Sales Manager	27000					
SA9	Mary Howe	Assistant	9000					
SL89	Stuart Stern	Secretary	8500	01/06/91	2350	SA2B	3700	100
SL31	Robert Chin	Snr Sales Asst	17000					
SG5	Susan Brand	Manager	24000					

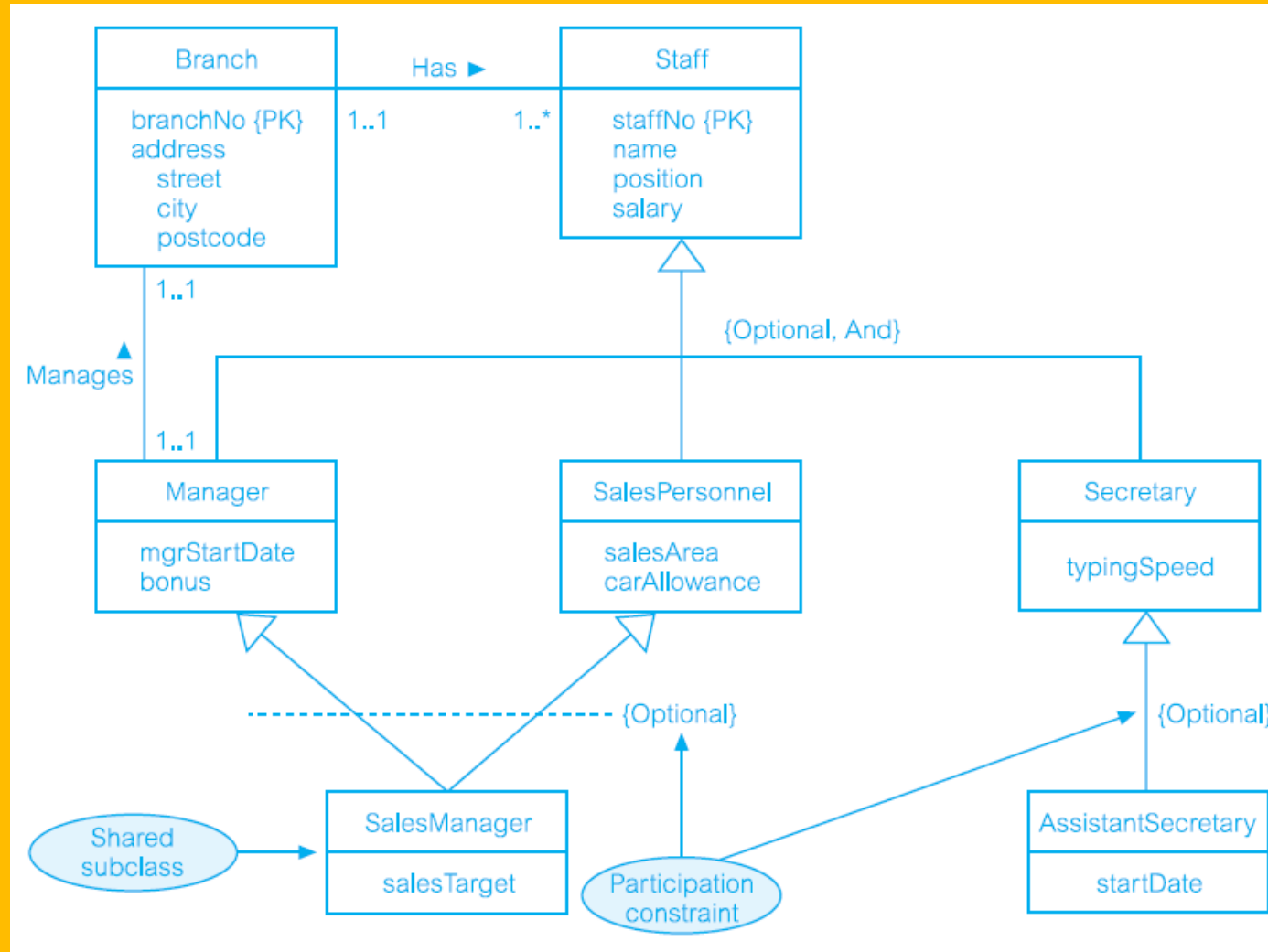
// X Specialization/generalization of Staff entity into subclasses representing job roles



Specialization/generalization of Staff entity into job roles and contracts of employment



EER diagram with shared subclass and subclass with its own subclass



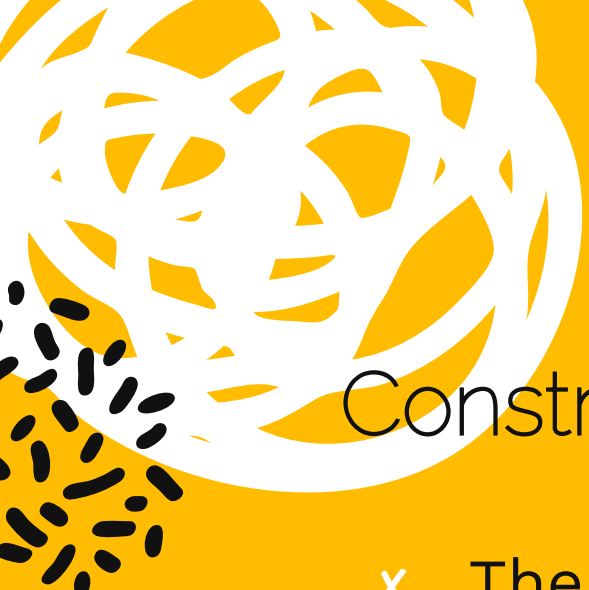
Constraints on Specialization / Generalization

Participation constraint

- Determines whether every member in superclass must participate as a member of a subclass.
- May be ***mandatory*** or ***optional***.

Disjoint constraint

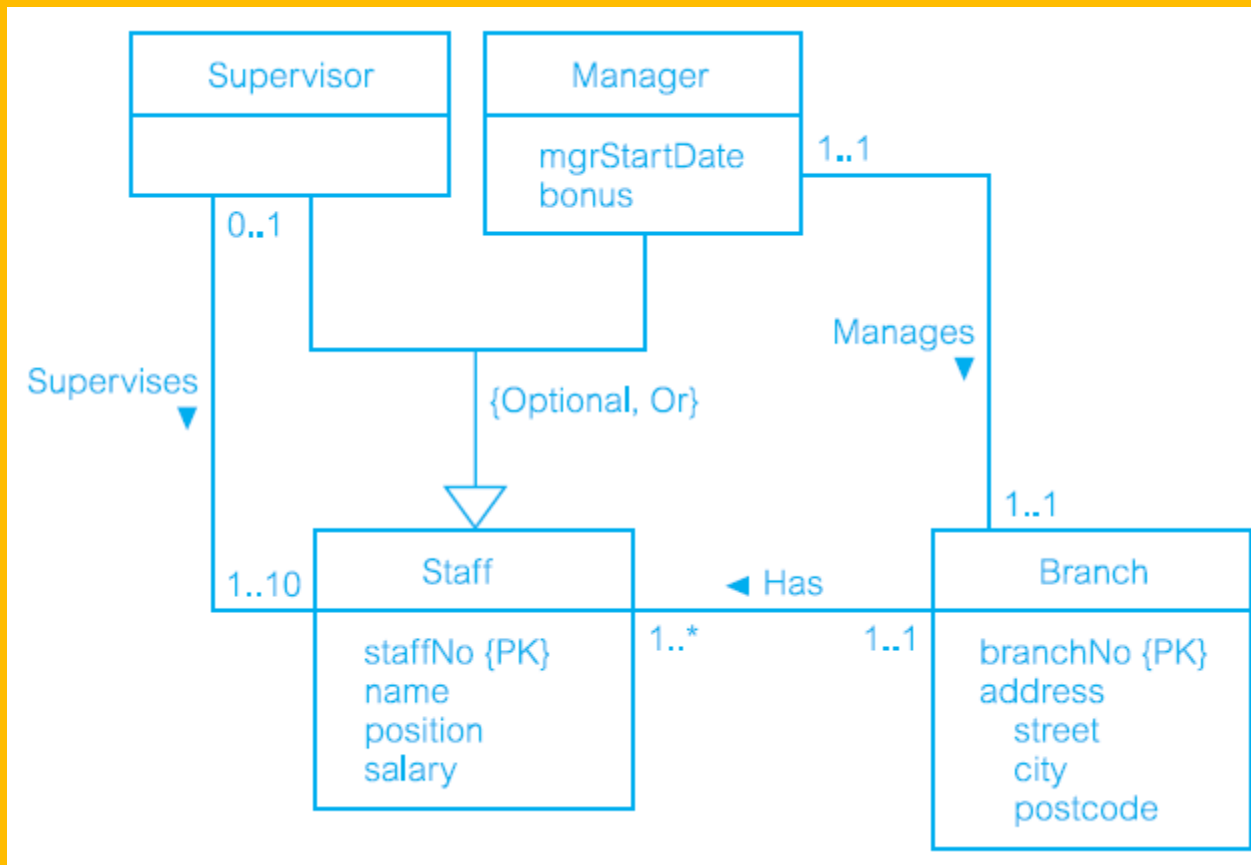
- Describes relationship between members of the subclasses and indicates whether member of a superclass can be a member of one, or more than one, subclass.
- May be ***disjoint*** or ***nondisjoint***.



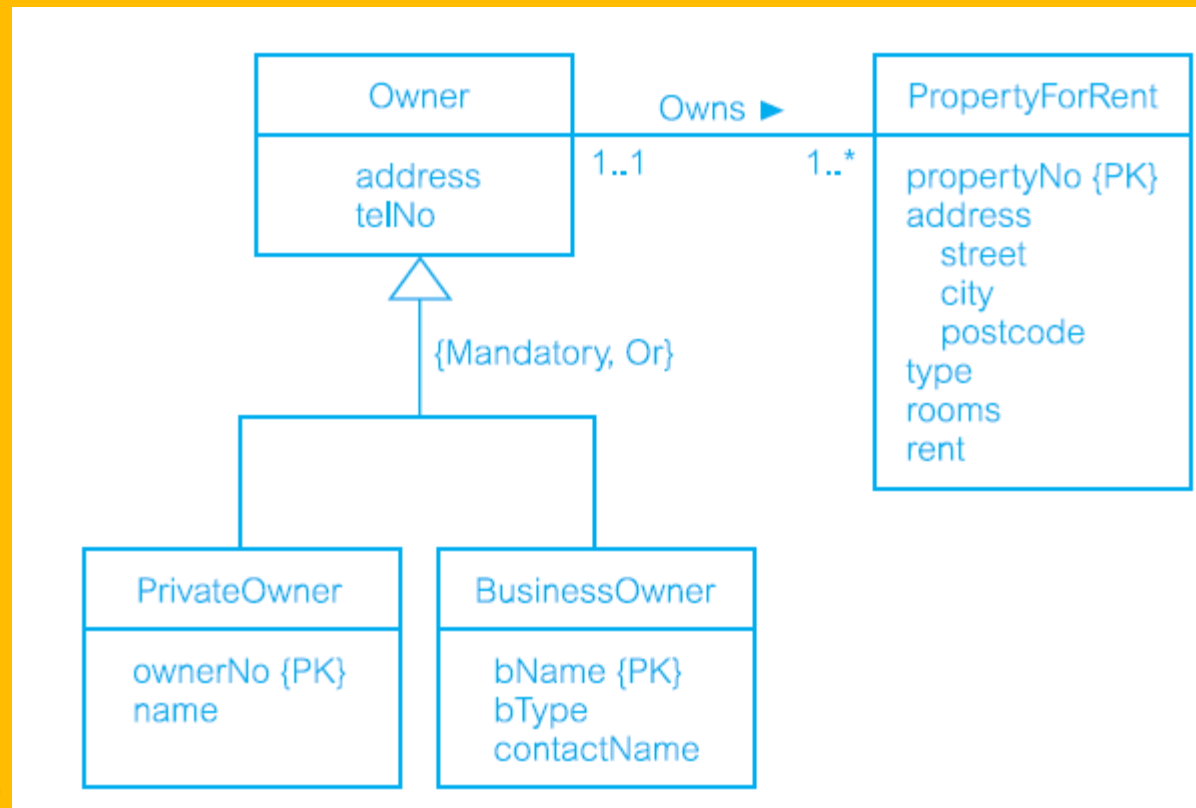
Constraints on Specialization / Generalization

- x There are four categories of constraints of specialization and generalization:
 - x mandatory and disjoint (or)
 - x optional and disjoint
 - x mandatory and nondisjoint (and) $\sqcap$ overlap
 - x optional and nondisjoint.
- 

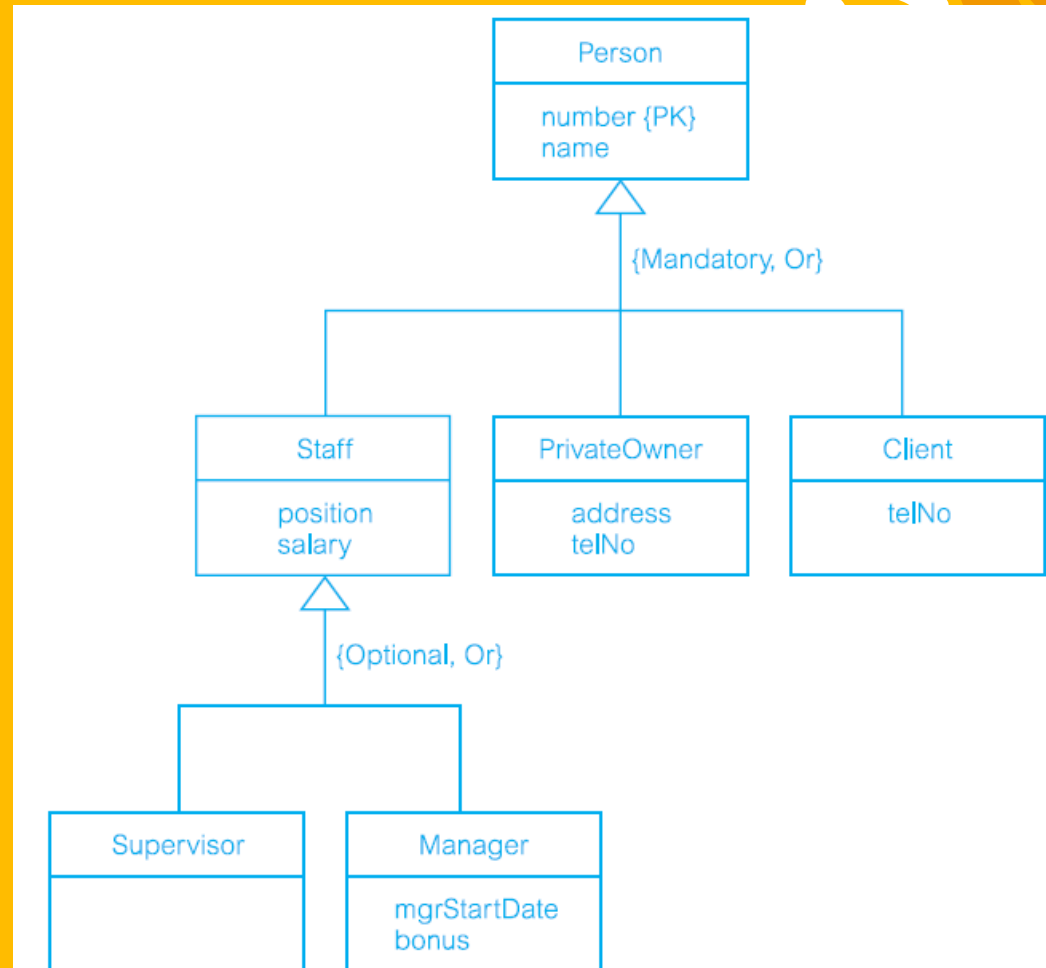
DreamHome worked example - Staff Superclass with Supervisor and Manager subclasses



DreamHome worked example - Owner Superclass with PrivateOwner and BusinessOwner subclasses



DreamHome worked example - Person superclass with Staff, PrivateOwner, and Client subclasses





References

- x Thomas Connolly and Carolyn Begg, Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition, Pearson, 2015, ISBN: 978-01329432
- 